EN.553.724 Probabilistic Machine Learning

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Chapter 1

Introduction

1.1 Probabilistic Modeling and Learning

1.1.1 Classical vs. Modern Methods

Classical probabilistic models (e.g. graphical models) and modern "deep" generative models differ primarily in how structure is specified.

- **Classical approaches**
  - Leverage domain knowledge to specify model structure.
  - Work well when the model is correctly specified.
  - Provide interpretable probabilistic quantities and uncertainty quantification.
- **Modern approaches**
  - Learn highly complex distributions directly from data.
  - Require large datasets.
  - Often lack interpretability.
- **Goal**

  combine structured probabilistic modeling with flexible function approximation.

1.1.2 Learning a Function

Model:

$$y = f(x) + \varepsilon, \quad \mathbb{E}[\varepsilon|x] = 0.$$

Given data $\{(x_i, y_i)\}_{i=1}^N$, choose f_θ.

$$\hat{\theta} = \arg\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} L(f_\theta(x_i), y_i).$$

1.1.3 Maximum Likelihood Interpretation

Assume:

$$\varepsilon \sim \mathcal{N}(0, I).$$

Then:

$$p(y_i|x_i, \theta) \propto \exp\left(-\frac{1}{2} \|y_i - f_\theta(x_i)\|^2\right).$$

Proposition 1.1.1. *Minimizing squared loss is equivalent to maximum likelihood estimation.*

$$\arg\max_{\theta} p(y_{1:N}|x_{1:N}) = \arg\min_{\theta} \sum_{i=1}^{N} \|y_i - f_\theta(x_i)\|^2.$$

1.1.4 Learning a Distribution

Given $x_1, \ldots, x_N \overset{\text{iid}}{\sim} \mathbb{P}$:

$$\hat{\theta} = \arg\max_{\theta} \sum_{i=1}^{N} \log p_\theta(x_i) = \arg\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} -\log p_\theta(x_i).$$

1.2 Measure-Theoretic Foundations

1.2.1 Discrete and Continuous Probability

- Discrete:
$$\mathbb{P}(A) = \sum_{x \in A} p(x)$$
- Continuous:
$$\mathbb{P}(A) = \int_A p(x)\ \mathrm{d}x$$

Unified via:

$$\mathbb{P}(A) = \int_A p(x)\, \lambda(\mathrm{d}x).$$

1.2.2 Probability Spaces

Definition 1.2.1. A measurable space is $(\Omega, \mathcal{F})$.

Definition 1.2.2. A probability measure $\mathbb{P}$ satisfies:

- $\mathbb{P}(A) \geq 0$
- $\mathbb{P}(\Omega) = 1$
- countable additivity

If a density exists:

$$\mathbb{P}(A) = \int_A p(x)\, \lambda(\,\mathrm{d}x).$$

1.2.3 Expectation

$$\mathbb{E}[f] = \int f(x)\, \mathbb{P}(\,\mathrm{d}x)$$

If density exists:

$$\mathbb{E}[f] = \int f(x)p(x)\, \lambda(\,\mathrm{d}x).$$

1.2.4 Radon–Nikodym Derivative

Definition 1.2.3. $\mathbb{P} \ll Q$ if $Q(A) = 0 \Rightarrow \mathbb{P}(A) = 0$.

Theorem 1.2.1 (Radon–Nikodym)**.** *If* $\mathbb{P} \ll Q$*, then*

$$\int f \,\mathrm{d}\mathbb{P} = \int f \frac{\mathrm{d}\mathbb{P}}{\mathrm{d}Q}\,\mathrm{d}Q.$$

If densities exist:

$$\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}Q}(x) = \frac{p(x)}{q(x)}.$$

1.2.5 Random Variables

Definition 1.2.4. A random variable is a measurable function $X : \Omega \to \mathbb{R}$.

Distribution:

$$\mathbb{P}_X(A) = \mathbb{P}(X \in A).$$

1.2.6 Conditional Probability

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

$$p(x|y) = \frac{p(x, y)}{p(y)}$$

$$p(x|y) = \frac{p(x)p(y|x)}{p(y)}$$

1.2.7 Beta-Binomial Model

$$\theta \sim \text{Beta}(\alpha, \beta), \quad X|\theta \sim \text{Binomial}(m+n, \theta).$$

Posterior:

$$\theta|X \sim \text{Beta}(\alpha + m, \beta + n).$$

1.2.8 Conditional Independence

Definition 1.2.5. $X \perp Y|Z$ if

$$p(x, y|z) = p(x|z)p(y|z).$$

1.3 Divergences and Information

1.3.1 Total Variation

Definition 1.3.1.

$$\text{TV}(\mathbb{P}, Q) = \frac{1}{2}\int |p - q| \, \lambda(\, \mathrm{d}x).$$

Proposition 1.3.1.

$$\text{TV}(\mathbb{P}, Q) = \max_A |\mathbb{P}(A) - Q(A)|.$$

1.3.2 f-Divergence

Definition 1.3.2.

$$D_f(\mathbb{P}\|Q) = \mathbb{E}_Q\left[f\left(\frac{p}{q}\right)\right].$$

$$\text{KL}(\mathbb{P}\|Q) = \int p \log \frac{p}{q}.$$

1.3.3 Non-Negativity

Theorem 1.3.1 (Jensen)**.**

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]).$$

Proposition 1.3.2.

$$D_f(\mathbb{P}\|Q) \geq 0.$$

Proof.

$$\mathbb{E}_Q f\left(\frac{p}{q}\right) \geq f\left(\mathbb{E}_Q \frac{p}{q}\right) = f(1) = 0.$$

□

1.3.4 Entropy and KL

Definition 1.3.3.

$$H(\mathbb{P}) = -\int p \log p.$$

$$\mathrm{KL}(\mathbb{P}\|Q) = -H(\mathbb{P}) - \mathbb{E}_{\mathbb{P}}[\log q].$$

1.3.5 MLE and KL

Proposition 1.3.3.

$$\mathbb{E}_{\mathbb{P}}[-\log p_\theta(x)] = \mathrm{KL}(\mathbb{P}\|\mathbb{P}_\theta) + H(\mathbb{P}).$$

$$\Rightarrow \mathrm{MLE} = \arg\min_\theta \mathrm{KL}(\mathbb{P}\|\mathbb{P}_\theta).$$

1.3.6 KL Direction

$$\mathrm{KL}(\mathbb{P}\|Q) \quad \text{mass-covering}$$

$$\mathrm{KL}(Q\|\mathbb{P}) \quad \text{mode-seeking}$$

1.3.7 Invariance

Proposition 1.3.4. *For invertible g:*

$$D_f(\mathbb{P}\|Q) = D_f(g_\#\mathbb{P}\|g_\#Q).$$

1.4 Optimal Transport and Integral Probability Metrics

1.4.1 Couplings and Wasserstein Distance

Definition 1.4.1. Let $(\Omega, \mathcal{F})$ be a measurable space with metric d, and let μ, ν be probability measures on Ω.

A *coupling* of μ and ν is a probability measure γ on $\Omega \times \Omega$ such that:

$$\gamma(A \times \Omega) = \mu(A), \quad \gamma(\Omega \times A) = \nu(A).$$

Interpretation: a coupling specifies a joint distribution (X, Y) with marginals $X \sim \mu$, $Y \sim \nu$.

Definition 1.4.2. The Wasserstein-p distance is defined as

$$W_p(\mu, \nu) = \left(\inf_{\gamma \in \Gamma(\mu,\nu)} \mathbb{E}_{(X,Y)\sim\gamma}[d(X,Y)^p] \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ is the set of all couplings of μ, ν.

Remark 1.4.1. W_1 is often called the *earth mover's distance*, since it measures the minimal cost of transporting mass from μ to ν.

1.4.2 Integral Probability Metrics

Definition 1.4.3. Let $\mathcal{F}$ be a class of functions $f : \Omega \to \mathbb{R}$. The integral probability metric (IPM) is

$$d_{\mathcal{F}}(\nu, \mu) = \sup_{f \in \mathcal{F}} \left| \mathbb{E}_\nu[f] - \mathbb{E}_\mu[f] \right|.$$

Lemma 1.4.1. *Let $\mathcal{F} = \{f : \|f\|_\infty \leq 1\}$. Then*

$$\mathrm{TV}(\mu, \nu) = \frac{1}{2} d_{\mathcal{F}}(\nu, \mu).$$

Proof. For any f with $\|f\|_\infty \leq 1$,

$$|\mathbb{E}_\nu[f] - \mathbb{E}_\mu[f]| = \left| \int f(x)(\nu - \mu)(\,\mathrm{d}x) \right| \leq \int |f(x)|\, |\nu - \mu|(\,\mathrm{d}x) \leq \int |\nu - \mu|(\,\mathrm{d}x).$$

Taking the supremum gives

$$d_{\mathcal{F}}(\nu, \mu) \leq 2\,\mathrm{TV}(\mu, \nu).$$

Equality is achieved by choosing

$$f(x) = \mathrm{sign}(p(x) - q(x)),$$

which yields

$$\mathbb{E}_\nu[f] - \mathbb{E}_\mu[f] = \int |p(x) - q(x)|\,\mathrm{d}x.$$

□

1.4.3 Kantorovich–Rubinstein Duality

Theorem 1.4.1 (Kantorovich–Rubinstein)**.** *Let $\mathcal{F}$ be the set of 1-Lipschitz functions on $\mathbb{R}^d$. Then*

$$W_1(\mu, \nu) = d_{\mathcal{F}}(\nu, \mu).$$

Sketch. For any coupling γ and any 1-Lipschitz f,

$$|f(x) - f(y)| \leq d(x, y).$$

Thus

$$\mathbb{E}_\nu[f] - \mathbb{E}_\mu[f] = \mathbb{E}_\gamma[f(Y) - f(X)] \leq \mathbb{E}_\gamma[d(X, Y)].$$

Taking infimum over γ gives

$$d_{\mathcal{F}}(\nu, \mu) \leq W_1(\mu, \nu).$$

The reverse inequality follows by constructing an optimal dual potential. □

Remarks on Metrics vs Divergences

- Total variation and Wasserstein are *metrics*.
- f-divergences (e.g. KL) are not metrics:
  - not symmetric
  - do not satisfy triangle inequality
- IPMs provide a unifying view of distances via function classes.

1.5 Maximum Entropy and Exponential Families

1.5.1 Maximum Entropy Principle

Let P be an unknown distribution on Ω. Suppose we are given moment constraints:

$$\mathbb{E}_P[\phi_i(X)] = \mu_i, \quad i = 1, \dots, d,$$

or equivalently

$$\mathbb{E}_P[\phi(X)] = \mu \in \mathbb{R}^d,$$

where $\phi : \Omega \to \mathbb{R}^d$.

Remark 1.5.1. These constraints alone do not uniquely determine P.

A natural principle is to choose the *least informative* distribution satisfying the constraints.

Definition 1.5.1. The maximum entropy distribution (with respect to a reference measure ν) is

$$\arg\max_P \; H_\nu(P) \quad \text{s.t.} \quad \mathbb{E}_P[\phi] = \mu,$$

where

$$H_\nu(P) = -\int p(x) \log p(x) \, \nu(\mathrm{d}x).$$

Derivation via Lagrange Multipliers

Assume Ω is finite for simplicity, and write $p(x) = P(X = x)$.

We maximize

$$-\sum_{x \in \Omega} p(x) \log p(x)$$

subject to:

$$\sum_x p(x) = 1, \quad \sum_x p(x)\phi_i(x) = \mu_i.$$

Define the Lagrangian:

$$\mathcal{L}(p, \lambda_0, \lambda) = -\sum_x p(x) \log p(x) + \lambda_0 \left(\sum_x p(x) - 1 \right) + \sum_{i=1}^d \lambda_i \left(\sum_x p(x)\phi_i(x) - \mu_i \right).$$

Taking derivatives:

$$\frac{\partial \mathcal{L}}{\partial p(x)} = -\log p(x) - 1 + \lambda_0 + \sum_{i=1}^{d} \lambda_i \phi_i(x).$$

Setting to zero:

$$\log p(x) = \lambda_0 - 1 + \sum_{i=1}^{d} \lambda_i \phi_i(x).$$

Thus

$$p(x) \propto \exp\left(\sum_{i=1}^{d} \lambda_i \phi_i(x)\right) = \exp(\langle \theta, \phi(x) \rangle),$$

where $\theta = (\lambda_1, \ldots, \lambda_d)$.

This gives the exponential family form.

1.5.2 Exponential Family

Definition 1.5.2. Let $(\mathcal{X}, \nu)$ be a measure space and $\phi : \mathcal{X} \to \mathbb{R}^d$.

The exponential family is

$$p_\theta(x) = \frac{\exp(\langle \theta, \phi(x) \rangle)}{Z(\theta)} = \exp\Big(\langle \theta, \phi(x) \rangle - A(\theta)\Big),$$

where

$$Z(\theta) = \int \exp(\langle \theta, \phi(x) \rangle)\, \nu(\,\mathrm{d}x), \quad A(\theta) = \log Z(\theta).$$

Remark 1.5.2.

- θ is the natural parameter
- $\phi(x)$ are sufficient statistics
- $A(\theta)$ is the log-partition function

1.5.2.1 Example: Gaussian from Exponential Family

Let $\phi(x) = (x, x^2)$ on $\mathbb{R}$.

Then

$$p_\theta(x) \propto \exp(\theta_1 x + \theta_2 x^2).$$

Completing the square:

$$\theta_1 x + \theta_2 x^2 = \theta_2 \left(x + \frac{\theta_1}{2\theta_2}\right)^2 - \frac{\theta_1^2}{4\theta_2}.$$

For $\theta_2 < 0$, this is a Gaussian:

$$p_\theta(x) = \mathcal{N}(\mu, \sigma^2), \quad \mu = -\frac{\theta_1}{2\theta_2}, \quad \sigma^2 = -\frac{1}{2\theta_2}.$$

1.5.3 Maximum Entropy Characterization

Theorem 1.5.1. *If there exists θ such that*

$$\mathbb{E}_{p_\theta}[\phi(X)] = \mu,$$

then p_θ is the maximum entropy distribution among all distributions satisfying the moment constraints.

Proof. Let P satisfy $\mathbb{E}_P[\phi] = \mu$. Then

$$\mathrm{KL}(P\|p_\theta) = \mathbb{E}_P[\log p(x) - \log p_\theta(x)] \geq 0.$$

Rearranging:

$$-H(P) \geq \mathbb{E}_P[\log p_\theta(x)].$$

Using exponential family form:

$$\mathbb{E}_P[\log p_\theta(x)] = \mathbb{E}_P[\langle\theta, \phi(x)\rangle - A(\theta)] = \langle\theta, \mu\rangle - A(\theta).$$

Thus

$$H(P) \leq A(\theta) - \langle\theta, \mu\rangle = H(p_\theta),$$

so p_θ maximizes entropy. □

1.5.4 Mean Parameter Mapping

Define the mapping

$$\nabla A(\theta) = \mathbb{E}_{p_\theta}[\phi(X)].$$

Definition 1.5.3. Let

$$\mathcal{M} = \{\mu \in \mathbb{R}^d : \exists P \text{ such that } \mathbb{E}_P[\phi(X)] = \mu\}$$

be the set of achievable mean parameters.

Definition 1.5.4. The statistics $\{\phi_i\}_{i=1}^d$ are *minimal* if

$$\{1, \phi_1, \ldots, \phi_d\} \text{ are linearly independent.}$$

Proposition 1.5.1. *If ϕ is minimal, then:*

- $\nabla A : \Theta \to \mathcal{M}$ *is injective*
- $\nabla A : \Theta \to \mathcal{M}^\circ$ *is surjective (onto the interior)*

Sketch. If ϕ is minimal, then $A(\theta)$ is strictly convex:

$$\nabla^2 A(\theta) = \mathrm{Cov}_{p_\theta}(\phi(X)) \succ 0.$$

Strict convexity implies injectivity of ∇A.

Surjectivity onto the interior follows from convex duality and properties of cumulant generating functions. □

1.5.5 Convexity and Geometry

Proposition 1.5.2. *The log-partition function satisfies:*

$$\nabla^2 A(\theta) = \mathrm{Cov}_{p_\theta}(\phi(X)) \succeq 0.$$

Remark 1.5.3.

- $A(\theta)$ is convex
- strictly convex if ϕ is minimal
- exponential families form a convex dual pair:

$$\theta \leftrightarrow \mu = \nabla A(\theta)$$

Non-Minimal Case

If ϕ is not minimal, then there exists $v \neq 0$ such that

$$\langle v, \phi(x) \rangle = c \quad \forall x.$$

Then

$$p_{\theta + tv}(x) = p_\theta(x),$$

so the parametrization is not identifiable, and ∇A is not injective.

Chapter 2

Graphical Models

2.1 Directed Graphical Models (Bayesian Networks)

2.1.1 Factorization in DAGs

Definition 2.1.1. Let $G = (V, E)$ be a directed acyclic graph (DAG), and let $\{X_v\}_{v \in V}$ be random variables.

A distribution P *factorizes according to* G if

$$p(x_V) = \prod_{v \in V} p\Big(x_v \mid x_{\mathrm{pa}(v)}\Big),$$

where $\mathrm{pa}(v)$ denotes the set of parents of node v.

Remark 2.1.1. Each factor $p(x_v \mid x_{\mathrm{pa}(v)})$ is a *local conditional model*.

Definition 2.1.2. A *topological ordering* of a DAG is an ordering $(v_1, \ldots, v_n)$ such that

$$v_i \to v_j \implies i < j.$$

Remark 2.1.2. This allows a generative interpretation:

$$X_{v_1} \to X_{v_2} \to \cdots \to X_{v_n},$$

sampling each variable conditioned on its parents.

2.1.2 Independence Structure

Definition 2.1.3. Let $S_v = \Big(\{v\} \cup \mathrm{Desc}(v) \cup \mathrm{pa}(v)\Big)^c$.

A DAG encodes the independence relations

$$X_v \perp X_{S_v} \mid X_{\mathrm{pa}(v)}.$$

Proposition 2.1.1. *A distribution factorizes according to G if and only if it satisfies the independence relations encoded by G.*

2.1.3 Factorization ⇒ Independence (Key Idea)

Lemma 2.1.1. *If P factorizes according to a DAG G, then for any upward-closed set U,*

$$p(x_U) = \prod_{u \in U} p(x_u \mid x_{\mathrm{pa}(u)}).$$

Sketch. Sum out variables not in U using the factorization; terms outside U cancel due to normalization. □

This leads to conditional independence:

$$X_v \perp X_{S_v} \mid X_{\mathrm{pa}(v)}.$$

2.1.4 Constructing a DAG

Given variables $X_1, \ldots, X_n$:

- Choose an ordering $X_1, \ldots, X_n$
- For each k, choose a minimal set $S \subseteq \{1, \ldots, k-1\}$ such that

$$X_k \perp X_{[k-1] \setminus S} \mid X_S$$

- Set $\mathrm{pa}(k) = S$

Remark 2.1.3. The resulting graph is not unique.

Remark 2.1.4. Causality can guide graph construction, but cannot generally be inferred from the graph alone.

2.1.5 Compactness of Representation

For n variables taking values in $\{1, \ldots, r\}$:

- Full joint distribution requires:

$$r^n - 1 \quad \text{parameters}$$

- If max indegree is d:

$$O(nr^{d+1})$$

  parameters suffice

Remark 2.1.5. Graphical models exploit conditional independence for exponential savings.

2.1.6 Examples

2.1.6.1 Markov Chain

$$X_{k+1} \perp X_1, \ldots, X_{k-1} \mid X_k$$

Factorization:

$$p(x_1, \ldots, x_n) = p(x_1) \prod_{k=2}^{n} p(x_k \mid x_{k-1})$$

2.1.6.2 d-th Order Markov Model

$$X_{k+d} \perp X_1, \ldots, X_{k-1} \mid X_k, \ldots, X_{k+d-1}$$

2.1.6.3 Naive Bayes

Latent class C:

$$X_k \perp X_{-k} \mid C$$

Factorization:

$$p(c, x_1, \ldots, x_n) = p(c) \prod_{k=1}^{n} p(x_k \mid c)$$

2.1.7 Latent Variable Models

Definition 2.1.4. A *latent variable model* introduces hidden variables Z such that

$$p(x) = \int p(x, z) \, \mathrm{d}z.$$

Remark 2.1.6. Latent variables simplify structure by introducing conditional independence.

2.1.7.1 Mixture Model

$$z_i \sim \text{Multinomial}(\pi), \quad x_i \mid z_i \sim \mathcal{N}(\mu_{z_i}, \Sigma_{z_i})$$

2.1.7.2 Stochastic Block Model

$$z_i \sim \text{Multinomial}(\pi), \quad A_{ij} \mid z_i, z_j \sim \text{Bernoulli}(p_{z_i z_j})$$

2.1.7.3 Noisy-OR Model

$$x_j \mid z \sim \text{Bernoulli}\left(1 - \exp\left(-\sum_i w_{ij} z_i\right)\right)$$

2.1.7.4 Hidden Markov Model (HMM)

$$z_t = Az_{t-1} + \epsilon_t, \quad x_t = Cz_t + \eta_t$$

Remark 2.1.7. This is a linear Gaussian state-space model.

2.1.8 d-Separation

Definition 2.1.5. A path in a DAG is *blocked* by a set C if one of the following holds:

1. Chain: $X \to Z \to Y$ or $X \leftarrow Z \leftarrow Y$, and $Z \in C$
2. Fork: $X \leftarrow Z \to Y$, and $Z \in C$
3. Collider: $X \to Z \leftarrow Y$, and $Z \notin C$ and no descendant of Z is in C

Definition 2.1.6. A and B are *d-separated* by C if all paths between them are blocked.

Theorem 2.1.1. *If P factorizes according to G and A, B are d-separated by C, then*

$$X_A \perp X_B \mid X_C.$$

2.2 Undirected Graphical Models (Markov Random Fields)

2.2.1 Factorization

Definition 2.2.1. Let $G = (V, E)$ be an undirected graph. Let $\mathcal{C}$ be the set of maximal cliques.

A distribution factorizes over G if

$$p(x) \propto \exp(-E(x)), \quad E(x) = \sum_{C \in \mathcal{C}} \phi_C(x_C),$$

or equivalently

$$p(x) = \prod_{C \in \mathcal{C}} \psi_C(x_C).$$

Remark 2.2.1. $\psi_C = e^{-\phi_C}$ are called *clique potentials*.

2.2.2 Markov Properties

Definition 2.2.2. P satisfies:

- Global Markov property:

$$X_A \perp X_B \mid X_C \quad \text{if } C \text{ separates } A, B$$

- Local Markov property:
$$X_v \perp X_{V\setminus(\{v\}\cup N(v))} \mid X_{N(v)}$$
- Pairwise Markov property:
$$X_v \perp X_w \mid X_{V\setminus\{v,w\}}, \quad (v,w) \notin E$$

Remark 2.2.2. Under positivity conditions:
$$\text{global} \iff \text{local} \iff \text{pairwise}$$

2.2.3 Energy-Based Models

Definition 2.2.3. An energy-based model defines
$$p_\theta(x) \propto e^{-E_\theta(x)}.$$

Remark 2.2.3.
- Lower energy $\Rightarrow$ higher probability
- $E_\theta(x) = \langle \theta, \phi(x) \rangle$ recovers exponential families

2.2.4 Example: Ising Model

$$p(x) \propto \exp\left(\sum_i h_i x_i + \sum_{(i,j)\in E} J_{ij} x_i x_j\right), \quad x_i \in \{-1, 1\}$$

Conditional distribution:
$$p(x_i = 1 \mid x_{-i}) = \frac{\exp(h_i + \sum_{j\in N(i)} J_{ij} x_j)}{\exp(h_i + \sum_{j\in N(i)} J_{ij} x_j) + \exp(-h_i - \sum_{j\in N(i)} J_{ij} x_j)}.$$

Remark 2.2.4. Unlike DAGs, sampling is generally difficult due to the global partition function.

2.3 Additional Properties of Independence

Intersection Property

Proposition 2.3.1. *Suppose $p(x, y, z, w)$ has positive density with respect to a product measure. Then the* intersection property *holds:*
$$\begin{cases} X \perp Y \mid Z, W, \\ X \perp W \mid Z, Y \end{cases} \quad \Rightarrow \quad X \perp (Y, W) \mid Z.$$

Remark 2.3.1. This property does *not* hold in general without positivity.

Proposition 2.3.2. *If the intersection property holds, then the global, local, and pairwise Markov properties are equivalent.*

Hammersley–Clifford Theorem

Theorem 2.3.1 (Hammersley–Clifford)**.** *Let $G = (V, E)$ be an undirected graph. Suppose P has positive density.*

Then:

$$P \text{ satisfies the Markov properties} \iff P \text{ factorizes over } G.$$

Remark 2.3.2. Positivity is essential: without it, independence does not imply factorization.

Canonical Representation

Theorem 2.3.2. *Any strictly positive distribution can be written as*

$$p(x) = \exp\left(\sum_{S \subseteq V} \xi_S(x_S)\right),$$

where

$$\xi_S(x_S) = \sum_{Z \subseteq S} (-1)^{|S|-|Z|} \log p(x_Z, x_{S^c} = 0).$$

Remark 2.3.3.

- This is an inclusion–exclusion type expansion
- If P factorizes over G, then $\xi_S = 0$ for non-cliques S

2.4 Factor Graphs

Definition 2.4.1. A *factor graph* is a bipartite graph with:

- variable nodes V
- factor nodes F

and edges only between variables and factors.

A distribution factorizes as

$$p(x_V) = \prod_{a \in F} \varphi_a(x_{N(a)}),$$

where $N(a)$ are variables connected to factor a.

Remark 2.4.1. Factor graphs unify:

- DAG factorizations
- MRF factorizations

2.4.1 Example: Restricted Boltzmann Machine (RBM)

Let visible units $v \in \{0,1\}^{d_v}$ and hidden units $h \in \{0,1\}^{d_h}$.

$$p(v,h) \propto \exp\left(\langle v, Wh\rangle + \langle b, v\rangle + \langle a, h\rangle\right)$$

Remark 2.4.2. Key property:

$$p(h \mid v) = \prod_j p(h_j \mid v), \quad p(v \mid h) = \prod_i p(v_i \mid h)$$

Remark 2.4.3. This conditional independence enables efficient Gibbs sampling.

2.4.2 Example: Coding Theory (LDPC)

$$p(x,z) \propto \prod_i \mathbb{1}\{H_i z = 0\} \prod_j p(x_j \mid z_j)$$

Remark 2.4.4. Inference corresponds to belief propagation.

2.5 Conditional Random Fields

Definition 2.5.1. Let $(V \cup W, E)$ be an undirected graph.

A *conditional random field (CRF)* is a conditional distribution

$$P(X_W \mid X_V) \propto \prod_{C \in \mathcal{C}} \psi_C(x_C),$$

where cliques C involve variables in W.

Remark 2.5.1. CRFs model conditional structure without specifying a joint model over (V, W).

2.6 Directed $\to$ Undirected Conversion

Definition 2.6.1. The *moralized graph* $\mathcal{M}[G]$ of a DAG G is obtained by:

- adding edges between all pairs of parents of each node
- dropping edge directions

Proposition 2.6.1. *If P factorizes according to a DAG G, then it factorizes according to the undirected graph $\mathcal{M}[G]$.*

2.7 Independence via Moralization

Theorem 2.7.1. *Let G be a DAG and A, B, C disjoint sets. The following are equivalent:*

1. *A and B are d-separated by C in G*
2. *A and B are separated by C in the moralized ancestral graph*

Moreover, if P factorizes according to G, then

$$X_A \perp X_B \mid X_C.$$

Definition 2.7.1. The *ancestral graph* is the subgraph induced by A, B, C and their ancestors.

2.8 Markov Blanket

Definition 2.8.1. A set U is a *Markov blanket* for v if

$$X_v \perp X_{V \setminus (U \cup \{v\})} \mid X_U.$$

Proposition 2.8.1. 1. *In an undirected graph, $N(v)$ is a Markov blanket.*

2. *In a DAG, the Markov blanket is:*

$$\mathrm{pa}(v) \cup \mathrm{ch}(v) \cup \text{co-pa}(v).$$

Remark 2.8.1. The Markov blanket gives the minimal conditioning set needed to isolate a node.

2.9 Learning Graph Structure (Undirected Case)

Proposition 2.9.1. *Assume P is positive. Define graph G by:*

$$(v, w) \in E \iff X_v \not\perp X_w \mid X_{V \setminus \{v,w\}}.$$

Then G is the unique minimal graph for which P satisfies the pairwise Markov property.

Comparison: Directed vs Undirected Models

- Directed models:
  - capture causal / generative structure
  - sampling is straightforward

  - require ordering / domain knowledge

- Undirected models:
  - symmetric relationships
  - easier structure learning
  - inference often harder (partition function)

2.10 Inference Problems

2.10.1 Core Tasks

Given a probabilistic model $p_\theta(x)$:

- **Sampling:**
$$x \sim p_\theta(x)$$
- **Marginals:**
$$p_\theta(x_i) = \int p_\theta(x)\, \nu^{\otimes(n-1)}(dx_{-i})$$
- **MAP (maximum a posteriori):**
$$x^* = \arg\max_x p_\theta(x)$$
- **Partition function:**
$$Z(\theta) = \int \psi_\theta(x)\, \nu^{\otimes n}(dx)$$

Remark 2.10.1. These tasks are tightly coupled: computing one often requires solving the others.

Latent Variable Models

For joint models:
$$p_\theta(x, z) \propto \psi_\theta(x, z)$$

posterior inference requires:

$$p_\theta(z|x) = \frac{\psi_\theta(x, z)}{Z(\theta, x)}, \quad Z(\theta, x) = \int \psi_\theta(x, z)\, \nu_Z(dz)$$

- Sampling: $z \sim p_\theta(z|x)$
- Marginals: $p_\theta(z_i|x)$
- MAP: $\arg\max_z p_\theta(z|x)$

2.11 Computational Complexity

Complexity Classes

- $O(n)$, $O(n \log n)$, $O(n^c)$: polynomial time
- $O(2^n)$: exponential time

Definition 2.11.1. A *decision problem* outputs YES/NO.

- **P**: solvable in polynomial time
- **NP**: solutions verifiable in polynomial time
- **NP-complete**: hardest problems in NP
- **NP-hard**: at least as hard as NP-complete

Example 2.11.1. 3-SAT: given a Boolean formula, does a satisfying assignment exist?

Counting Problems

Definition 2.11.2. #P: counting versions of NP problems (e.g., number of satisfying assignments).

Remark 2.11.1. Computing partition functions and marginals is typically #P-hard.

Theorem 2.11.1. *Computing partition functions or marginals in general graphical models is #P-hard.*

Remark 2.11.2. Even approximate counting (e.g., #SAT) can be NP-hard.

2.12 Approximation Algorithms

Definition 2.12.1. A *polynomial-time randomized approximation scheme (PRAS)* for Z outputs $\widehat{Z}$ such that

$$e^{-\epsilon} Z \leq \widehat{Z} \leq e^{\epsilon} Z$$

in polynomial time, with high probability.

Definition 2.12.2.

- **FPRAS**: runtime polynomial in n and $1/\epsilon$
- **PTAS**: deterministic approximation scheme

Equivalence: Sampling, Marginals, Partition Function

Definition 2.12.3. A family of problems is *self-reducible* if fixing variables reduces the problem to a smaller instance.

Theorem 2.12.1 (informal)**.** *For distributions $p_\theta(x) \propto \psi_\theta(x)$ on finite spaces:*

1. *Approximate sampling $\Rightarrow$ approximate marginals*
2. *Approximate marginals $\Rightarrow$ approximate sampling (if self-reducible)*
3. *Marginals $\Leftrightarrow$ partition function*

Theorem 2.12.2. *If we can sample within TV distance ϵ in $\mathrm{poly}(n, 1/\epsilon)$ time, then we can approximate marginals via empirical averages:*

$$\widehat{p}(x_i = a) = \frac{1}{N} \sum_{k=1}^{N} \mathbb{1}\{x_i^{(k)} = a\}$$

Proof. By Hoeffding's inequality:

$$\mathbb{P}\left(|\widehat{p} - p| \geq \delta\right) \leq 2\exp(-2N\delta^2)$$

so $N = O(1/\delta^2)$ suffices. □

2.13 Exact Inference

2.13.1 Variable Elimination

Algorithm 1 Variable Elimination

Input: $p(x) \propto \prod_a \psi_a(x_{\partial a})$
Choose elimination order
for variable i **do**
    Multiply all factors involving i
    Sum out i
    Introduce new factor
end for
Output desired marginal

Remark 2.13.1. Intermediate factors grow in size $\rightarrow$ exponential complexity.

Treewidth

Definition 2.13.1. The *treewidth* $\mathrm{tw}(G)$ is the size of the largest intermediate factor (minus one) in optimal elimination ordering.

Remark 2.13.2. Complexity of variable elimination:

$$O(n \cdot \mathrm{state}^{\mathrm{tw}(G)+1})$$

2.14 Belief Propagation

2.14.1 Message Passing

Messages:

- Variable $\to$ factor:
$$m_{i\to a}(x_i) \propto \prod_{b\in\partial i\setminus a} m_{b\to i}(x_i)$$
- Factor $\to$ variable:
$$m_{a\to i}(x_i) \propto \sum_{x_{\partial a\setminus i}} \psi_a(x_{\partial a}) \prod_{j\in\partial a\setminus i} m_{j\to a}(x_j)$$

Marginals

$$p(x_i) \propto \prod_{a\in\partial i} m_{a\to i}(x_i)$$

$$p(x_{\partial a}) \propto \psi_a(x_{\partial a}) \prod_{j\in\partial a} m_{j\to a}(x_j)$$

2.14.2 Properties

- Exact on trees
- Equivalent to dynamic programming
- Linear-time in number of nodes (for trees)

Remark 2.14.1. On loopy graphs:

- becomes *loopy belief propagation*
- no guarantees, but often works well

2.15 Extensions

- Conditioning: add indicator factors
- Continuous variables: approximate messages (e.g., Gaussian)
- Junction tree algorithm: exact inference via tree decomposition

2.16 Takeaways

- Exact inference is exponential in general
- Tree structure enables efficient computation
- Message passing = dynamic programming on graphs
- Sampling, marginals, and partition function are deeply connected

Correctness of Belief Propagation (Trees)

We formalize what the messages compute.

Let $T_{b\to i}$ denote the subtree rooted at b when edge (b, i) is removed.

Proposition 2.16.1. *For a tree-structured factor graph:*

$$Z_{b\to i}(x_i) = \sum_{x_{V(T_{b\to i})\setminus i}} \prod_{c\in F(T_{b\to i})} \psi_c(x_{\partial c}),$$

and similarly

$$Z_{a\to j}(x_j) = \sum_{x_{V(T_{a\to j})\setminus j}} \prod_{c\in F(T_{a\to j})} \psi_c(x_{\partial c}).$$

Proof. By induction on the size of the subtree.

Base case: leaf node. Only one factor contributes, and the expression holds trivially.

Inductive step: assume the claim holds for all subtrees of depth $\leq k$. Then for a larger subtree,

$$Z_{a\to i}(x_i) = \sum_{x_{\partial a\setminus i}} \psi_a(x_{\partial a}) \prod_{j\in\partial a\setminus i} Z_{j\to a}(x_j),$$

which matches the factor-to-variable message update.

Thus the recursion exactly computes subtree marginals. □

Max-Product Belief Propagation (MAP)

Replace sums by maxima.

- Variable → factor:

$$M_{i\to a}(x_i) = \prod_{b\in\partial i\setminus a} M_{b\to i}(x_i)$$

- Factor → variable:

$$M_{a\to i}(x_i) = \max_{x_{\partial a\setminus i}} \psi_a(x_{\partial a}) \prod_{j\in\partial a\setminus i} M_{j\to a}(x_j)$$

Remark 2.16.1. This computes max-marginals:

$$M_i(x_i) \propto \max_{x_{-i}} p(x)$$

Algorithm 2 Max-product belief propagation

Initialize messages
repeat
  for factor a **do**
    update $M_{a \to i}$
  end for
  for variable i **do**
    update $M_{i \to a}$
  end for
until convergence
Output $\arg\max_x p(x)$ from max-marginals

2.16.1 Hidden Markov Models (HMMs)

Model

Latent chain:

$$z_1 \to z_2 \to \cdots \to z_n, \quad x_t | z_t$$

Factorization:

$$p(z_{1:n}, x_{1:n}) = p(z_1) \prod_{t=2}^{n} p(z_t | z_{t-1}) \prod_{t=1}^{n} p(x_t | z_t)$$

Forward Pass

Define

$$f_t(i) = p(z_t = i, x_{1:t})$$

Recursion:

$$f_{t+1}(j) = p(x_{t+1} | z_{t+1} = j) \sum_i f_t(i)\, p(z_{t+1} = j | z_t = i)$$

Backward Pass

$$b_t(i) = p(x_{t+1:n} | z_t = i)$$

Recursion:

$$b_t(i) = \sum_j p(z_{t+1} = j | z_t = i)\, p(x_{t+1} | z_{t+1} = j)\, b_{t+1}(j)$$

Posterior

$$p(z_t = i|x_{1:n}) \propto f_t(i)\, b_t(i)$$

Viterbi Algorithm (MAP)

Define

$$\delta_t(j) = \max_{z_{1:t-1}} p(z_{1:t}, x_{1:t})$$

Recursion:

$$\delta_{t+1}(j) = p(x_{t+1}|z_{t+1} = j) \max_i \delta_t(i)\, p(z_{t+1} = j|z_t = i)$$

Remark 2.16.2. Viterbi is max-product BP on a chain.

2.16.2 Loopy Belief Propagation

- Same updates as BP, but on graphs with cycles
- No guarantees of convergence or correctness
- Often works well in practice

Example 2.16.1. LDPC decoding uses loopy BP.

Summary: VE vs BP

- **Variable elimination:**
  - works on any graph
  - computes one marginal at a time
  - complexity depends on treewidth
- **Belief propagation:**
  - exact on trees
  - computes all marginals simultaneously
  - linear-time for trees
  - approximate on loopy graphs

2.17 Takeaways

- Exact inference is exponential in general
- Trees ⇒ efficient dynamic programming

- Message passing unifies many algorithms:
  - forward-backward
  - Viterbi
  - BP
- MAP inference = max-product BP

2.18 Sampling with Markov Chains

Problem Formulation

We want to approximately sample from a distribution

$$p(x) \propto e^{f(x)}, \quad x \in \Omega,$$

within ε in total variation distance.

Remark 2.18.1. This problem appears across:

- Bayesian inference
- statistical physics
- combinatorics / theoretical CS
- deep generative modeling

Remark 2.18.2. Sampling is essentially equivalent to computing expectations:

$$\mathbb{E}_{x\sim p}[f(x)] = \int f(x)p(x)\,dx,$$

which becomes intractable in high dimensions.

2.18.1 Monte Carlo Estimation

Given samples $X_1, \ldots, X_N \sim p$,

$$\mathbb{E}[f(X)] \approx \frac{1}{N}\sum_{i=1}^{N} f(X_i).$$

Proposition 2.18.1.

$$\mathrm{Var}\left(\frac{1}{N}\sum_{i=1}^{N} f(X_i)\right) = \frac{\mathrm{Var}(f(X))}{N}.$$

Proof. Uses independence:

$$\mathrm{Var}\Big(\frac{1}{N}\sum f(X_i)\Big) = \frac{1}{N^2}\sum \mathrm{Var}(f(X_i)) = \frac{1}{N}\mathrm{Var}(f(X)).$$

□

What Distributions Are Easy to Sample?

- Uniform distributions
- Finite discrete distributions
- 1D distributions via inverse CDF
- Product distributions:
$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p_i(x_i)$$
- Directed graphical models:
$$p(x) = \prod_{i=1}^{n} p(x_i | x_{\text{parents}(i)})$$
- Gaussian distributions

Remark 2.18.3. General high-dimensional distributions are not directly sampleable.

2.18.2 Importance Sampling

Let $Y \sim Q$.

Definition 2.18.1.
$$\widehat{\mu} = \frac{1}{N} \sum_{i=1}^{N} \frac{p(Y_i)}{q(Y_i)} f(Y_i).$$

Proposition 2.18.2.
$$\mathbb{E}_Q[\widehat{\mu}] = \mathbb{E}_P[f(X)].$$

Proof.
$$\mathbb{E}_Q\left[\frac{p(Y)}{q(Y)} f(Y)\right] = \int \frac{p(y)}{q(y)} f(y) q(y)\, dy = \int f(y) p(y)\, dy.$$

□

2.18.3 Rejection Sampling

Algorithm 3 Rejection sampling

Input: $q(x)$, c such that $p(x) \leq cq(x)$
repeat
  Sample $Y \sim Q$
  Sample $U \sim \text{Uniform}(0, 1)$
until $U \leq \frac{p(Y)}{cq(Y)}$
Output Y

Theorem 2.18.1. *If $p(x) \leq cq(x)$, then rejection sampling produces samples from p, and*

$$\#\textit{trials} \sim \text{Geom}(1/c).$$

Remark 2.18.4. Fails in high dimensions due to exponential scaling of c.

2.19 Markov Chains

Definition 2.19.1. A sequence (X_t) is a Markov chain if

$$X_{t+1} \perp X_1, \ldots, X_{t-1} \mid X_t,$$

i.e.,

$$p(x_{t+1}|x_1, \ldots, x_t) = p(x_{t+1}|x_t).$$

Definition 2.19.2. A Markov chain is time-homogeneous if

$$p(x_{t+1}|x_t) \text{ does not depend on } t.$$

Transition Kernels

Definition 2.19.3. A Markov kernel K is a function such that:

- $K(x, \cdot)$ is a probability measure
- measurable in x

For a distribution μ,

$$(\mu K)(A) = \int K(x, A)\, \mu(dx).$$

If densities exist:

$$(\mu K)(y) = \int \mu(x) k(x, y)\, dx.$$

Operator View

Definition 2.19.4. For $f : \Omega \to \mathbb{R}$,

$$Kf(x) = \mathbb{E}_{Y \sim K(x,\cdot)}[f(Y)] = \int f(y) K(x, dy).$$

Remark 2.19.1. K acts:

- on distributions: $\mu \mapsto \mu K$
- on functions: $f \mapsto Kf$

2.19.1 Stationary Distributions

Definition 2.19.5. π is stationary if

$$\pi K = \pi.$$

Remark 2.19.2. Then $\pi K^t = \pi$ for all t.

2.19.2 Conditions for Convergence

Definition 2.19.6. A Markov chain is:

- irreducible: any state can reach any other
- aperiodic: no deterministic cycles

Theorem 2.19.1. *If a finite Markov chain is irreducible and aperiodic:*

- *it has a unique stationary distribution* π
- $\mu_0 K^t \to \pi$

Theorem 2.19.2 (Ergodic theorem)**.**

$$\frac{1}{N}\sum_{t=1}^{N} f(X_t) \to \mathbb{E}_\pi[f(X)] \quad a.s.$$

2.19.3 Reversibility

Definition 2.19.7. π satisfies detailed balance if

$$\pi(x)K(x,y) = \pi(y)K(y,x).$$

Proposition 2.19.1. *Detailed balance* $\Rightarrow$ *stationarity.*

Remark 2.19.3. Reversibility makes it easy to design Markov chains.

2.20 Metropolis–Hastings

Target density: $p(x) \propto \tilde{p}(x)$.

Theorem 2.20.1. *The MH chain has stationary distribution* p.

Gibbs Sampling

Definition 2.20.1. At each step:

$$X_i \sim p(x_i | x_{-i}).$$

Algorithm 4 Metropolis–Hastings

Input: $\tilde{p}(x)$, proposal $q(x, y)$
for $t = 0, \ldots, T$ **do**
Sample proposal $\widehat{X}_{t+1} \sim q(X_t, \cdot)$
Accept with probability

$$\alpha(x, y) = \min \left\{ \frac{\tilde{p}(y)q(y, x)}{\tilde{p}(x)q(x, y)}, 1 \right\}$$

end for

Proposition 2.20.1. *Gibbs sampling is a special case of Metropolis–Hastings with acceptance probability* 1.

Remark 2.20.1. In graphical models:

$$p(x_i|x_{-i}) = p(x_i|x_{\mathrm{MB}(i)}).$$

Examples

Example 2.20.1 (Ball walk)**.**

$$\widehat{X}_{t+1} \sim \mathrm{Uniform}(B_R(X_t)),$$

then accept/reject via MH.

Example 2.20.2 (Gibbs sampling)**.** Select coordinate i and sample

$$X_i \sim p(x_i|x_{-i}).$$

Takeaways

- Direct sampling is rarely possible
- MCMC constructs a Markov chain with target stationary distribution
- Convergence requires irreducibility + aperiodicity
- Reversibility simplifies construction (MH)
- Gibbs exploits conditional structure

2.21 Markov Chain Monte Carlo (MCMC)

2.21.1 Sampling via Markov Chains

Definition 2.21.1. Goal: sample from a distribution

$$p(x) \propto e^{f(x)}, \quad x \in \Omega,$$

within ε in total variation distance.

Remark 2.21.1. We construct a Markov chain whose stationary distribution is p, then simulate it.

Rejection Sampling (Limitations)

Theorem 2.21.1. *If $p(x) \le cq(x)$, rejection sampling produces samples from p, and*

$$\#\textit{trials} \sim \mathrm{Geom}(1/c).$$

Remark 2.21.2. In high dimensions, c typically grows exponentially, making rejection sampling impractical.

Markov Chains

Definition 2.21.2. A sequence (X_t) is a Markov chain if

$$p(x_{t+1}|x_1, \dots, x_t) = p(x_{t+1}|x_t).$$

Definition 2.21.3. A distribution π is stationary if

$$\pi K = \pi.$$

2.21.2 Reversibility

Definition 2.21.4. π satisfies detailed balance if

$$\pi(x)K(x,y) = \pi(y)K(y,x).$$

Proposition 2.21.1. *If detailed balance holds, then π is stationary.*

Proof.

$$\sum_x \pi(x)K(x,y) = \sum_x \pi(y)K(y,x) = \pi(y)\sum_x K(y,x) = \pi(y).$$

□

Metropolis–Hastings

Theorem 2.21.2. *The Metropolis–Hastings chain has stationary distribution p.*

Proof. Check detailed balance:

$$p(x)K(x,y) = p(x)q(x,y)\alpha(x,y) = \min\{p(x)q(x,y), p(y)q(y,x)\},$$

which is symmetric in (x,y). □

Algorithm 5 Metropolis–Hastings

Input: $\tilde{p}(x)$, proposal $q(x, y)$
for $t = 0, \dots, T$ **do**
Sample $Y \sim q(X_t, \cdot)$
Accept with probability

$$\alpha(x, y) = \min\left\{\frac{\tilde{p}(y)q(y, x)}{\tilde{p}(x)q(x, y)}, 1\right\}$$

Set $X_{t+1} = Y$ if accepted, else $X_{t+1} = X_t$
end for

Gibbs Sampling

Definition 2.21.5. At each step, pick coordinate i and sample

$$X_i \sim p(x_i|x_{-i}).$$

Proposition 2.21.2. *Gibbs sampling is a special case of Metropolis–Hastings with acceptance probability* 1.

Remark 2.21.3. In graphical models:

$$p(x_i|x_{-i}) = p(x_i|x_{\mathrm{MB}(i)}).$$

2.22 Langevin Monte Carlo (LMC)

Continuous-time process

Definition 2.22.1. Langevin diffusion:

$$dX_t = -\nabla V(X_t)\, dt + \sqrt{2}\, dB_t.$$

Remark 2.22.1. Target distribution:

$$p(x) \propto e^{-V(x)}.$$

Discretization

Using Euler–Maruyama:

$$X_{t+1} = X_t - \eta \nabla V(X_t) + \sqrt{2\eta}\, \xi_t, \quad \xi_t \sim \mathcal{N}(0, I).$$

Remark 2.22.2. This is noisy gradient descent.

Stationary Distribution

Theorem 2.22.1. *The Langevin diffusion has stationary distribution*

$$p(x) \propto e^{-V(x)}.$$

Proof. From the Fokker–Planck equation:

$$\frac{\partial \mu_t}{\partial t} = \nabla \cdot (\nabla V \, \mu_t) + \Delta \mu_t.$$

Plug $\mu(x) = e^{-V(x)}$:

$$\nabla \mu = -(\nabla V) e^{-V}, \quad \Delta \mu = \left(-\Delta V + \|\nabla V\|^2\right) e^{-V}.$$

Then:

$$\nabla \cdot (\nabla V \mu) = (\Delta V) e^{-V} - \|\nabla V\|^2 e^{-V}.$$

Adding:

$$\nabla \cdot (\nabla V \mu) + \Delta \mu = 0.$$

Thus μ is stationary. □

2.22.1 MALA (Metropolis-adjusted Langevin)

Algorithm 6 MALA

Propose

$$Y \sim \mathcal{N}(x - h\nabla V(x), 2hI)$$

Accept with MH correction

Remark 2.22.3. Corrects discretization bias of LMC.

2.23 Underdamped Langevin Dynamics

Definition 2.23.1. Introduce momentum v:

$$\begin{aligned} dX_t &= v_t \, dt \\ dv_t &= -\gamma v_t \, dt - \nabla V(X_t) \, dt + \sqrt{2\gamma} \, dB_t \end{aligned}$$

Remark 2.23.1. Stationary distribution:

$$p(x, v) \propto e^{-V(x) - \frac{1}{2}\|v\|^2}.$$

Remark 2.23.2. Non-reversible → faster mixing.

Hamiltonian Monte Carlo (HMC)

Hamiltonian dynamics

Define:

$$H(q,p) = U(q) + K(p)$$

$$\frac{dq}{dt} = \nabla_p H = \nabla K(p)$$
$$\frac{dp}{dt} = -\nabla_q H = -\nabla U(q)$$

Proposition 2.23.1. *Hamiltonian dynamics preserve:*

- *Energy:* $H(q(t), p(t))$
- *Volume (Liouville's theorem)*

Reversibility trick

Define momentum flip:

$$R(q,p) = (q,-p).$$

Then:

$$R \circ S_T \circ R = S_T^{-1}.$$

Proposition 2.23.2. *HMC transition is reversible w.r.t.*

$$\mu(q,p) \propto e^{-H(q,p)}.$$

Algorithm

Algorithm 7 Hamiltonian Monte Carlo (idealized)

Sample $p \sim e^{-K(p)}$
Simulate Hamiltonian dynamics for time T
Output q

Remark 2.23.3. No rejection needed in exact dynamics (but required after discretization).

Leapfrog Integrator

Remark 2.23.4. Properties:

- volume-preserving
- reversible
- approximately energy-preserving

Algorithm 8 Leapfrog

$p \leftarrow p - \frac{h}{2}\nabla U(q)$
for $t = 1, \ldots, T$ **do**
  $q \leftarrow q + h\nabla K(p)$
  $p \leftarrow p - h\nabla U(q)$
end for
$p \leftarrow p - \frac{h}{2}\nabla U(q)$

Takeaways

- MCMC constructs Markov chains with target stationary distribution
- Metropolis–Hastings ensures correctness via detailed balance
- Gibbs exploits conditional structure
- Langevin adds gradient information (diffusion-based sampling)
- HMC uses Hamiltonian dynamics for efficient exploration

Langevin Monte Carlo

We want to sample from

$$p(x) \propto e^{-V(x)}.$$

Definition 2.23.2. The Langevin diffusion is the stochastic differential equation

$$dX_t = -\nabla V(X_t)\, dt + \sqrt{2}\, dB_t.$$

Discretizing via Euler–Maruyama:

$$X_{t+1} = X_t - \eta\nabla V(X_t) + \sqrt{2\eta}\,\xi_t, \quad \xi_t \sim \mathcal{N}(0, I_d).$$

Remark 2.23.5. This can be viewed as gradient descent with injected Gaussian noise.

Remark 2.23.6. The continuous-time process has stationary distribution $p(x) \propto e^{-V(x)}$, but the discretized version introduces bias unless η is small.

Metropolis-Adjusted Langevin Algorithm (MALA)

To correct discretization bias, use Langevin as a proposal in Metropolis–Hastings.

Definition 2.23.3. Proposal distribution:

$$q(x, y) \propto \exp\left(-\frac{\|y - (x - \eta\nabla V(x))\|^2}{4\eta}\right).$$

Algorithm 9 MALA

Propose $Y \sim \mathcal{N}(x - \eta \nabla V(x),\, 2\eta I)$
Accept with probability

$$\alpha(x, y) = \min\left\{\frac{p(y)q(y, x)}{p(x)q(x, y)}, 1\right\}$$

Underdamped Langevin Dynamics

Definition 2.23.4. Introduce momentum v:

$$\begin{aligned} dX_t &= v_t\, dt, \\ dv_t &= -\gamma v_t\, dt - \nabla V(X_t)\, dt + \sqrt{2\gamma}\, dB_t. \end{aligned}$$

Remark 2.23.7. Stationary distribution:

$$p(x, v) \propto e^{-V(x) - \frac{1}{2}\|v\|^2}.$$

Remark 2.23.8. Noise is injected only in v_t, leading to more efficient exploration.

Continuous-Time Markov Processes

Definition 2.23.5. A continuous-time Markov process $(X_t)_{t \geq 0}$ satisfies

$$P(X_u \in A | X_{[0,t]}) = P(X_u \in A | X_t), \quad u > t.$$

Definition 2.23.6. A family $(P_t)_{t \geq 0}$ is a Markov semigroup if

$$P_{t+s} = P_t P_s, \quad P_0 = \mathrm{id}.$$

Generator of a Markov Process

Definition 2.23.7. The generator $\mathcal{L}$ is

$$\mathcal{L} f = \left| \frac{d}{dt} P_t f \right|_{t=0}.$$

Remark 2.23.9. This defines the process via

$$P_t = e^{t\mathcal{L}}.$$

2.23.1 Fokker–Planck Equation

Let $\mathcal{L}^*$ be the adjoint operator.

Definition 2.23.8. The distribution μ_t evolves as

$$\frac{d\mu_t}{dt} = \mathcal{L}^* \mu_t.$$

Remark 2.23.10. For Langevin diffusion:

$$\mathcal{L}f = -\langle \nabla f, \nabla V \rangle + \Delta f,$$
$$\mathcal{L}^* \mu = \nabla \cdot (\nabla V \mu) + \Delta \mu.$$

Stationarity of Langevin

Theorem 2.23.1. *The distribution*

$$\mu(x) \propto e^{-V(x)}$$

is stationary for Langevin diffusion.

Proof. Plug into Fokker–Planck equation:

$$\mathcal{L}^* \mu = \nabla \cdot (\nabla V \mu) + \Delta \mu.$$

Compute:

$$\nabla \mu = -(\nabla V)\mu, \quad \Delta \mu = (-\Delta V + \|\nabla V\|^2)\mu.$$

Then:

$$\nabla \cdot (\nabla V \mu) = (\Delta V)\mu - \|\nabla V\|^2 \mu.$$

Adding:

$$(\Delta V - \|\nabla V\|^2)\mu + (-\Delta V + \|\nabla V\|^2)\mu = 0.$$

□

2.24 Hamiltonian Monte Carlo (HMC)

2.24.1 Hamiltonian Dynamics

Define

$$H(q,p) = U(q) + K(p).$$

Dynamics:

$$\frac{dq}{dt} = \nabla_p H, \quad \frac{dp}{dt} = -\nabla_q H.$$

Remark 2.24.1. Typical choice:

$$K(p) = \frac{1}{2}\|p\|^2.$$

2.24.2 Key Properties

Proposition 2.24.1. *Hamiltonian dynamics preserve:*

- *energy $H(q, p)$*
- *volume in phase space*

2.24.3 Leapfrog Integrator

Lemma 2.24.1. *The leapfrog integrator:*

- *preserves volume*
- *is reversible (up to momentum flip)*

Remark 2.24.2. It does *not* exactly preserve the Hamiltonian.

HMC with Leapfrog

Algorithm 10 Hamiltonian Monte Carlo (practical)

Sample $p \sim e^{-K(p)}$
Simulate Hamiltonian dynamics via leapfrog for time T
Accept with probability

$$\alpha = \min\{e^{-H(q',p')+H(q,p)}, 1\}$$

Remark 2.24.3. The Metropolis step corrects discretization error.

Choosing Parameters

- Integration time T:
  - Too small $\Rightarrow$ slow exploration
  - Too large $\Rightarrow$ wasted computation
- Randomizing T helps avoid periodic behavior
- No-U-Turn criterion adaptively selects trajectory length

Geometric Adaptation

Definition 2.24.1. Riemannian HMC uses position-dependent metric:

$$K(q, p) = \frac{1}{2} p^\top \Sigma^{-1}(q) p + \frac{1}{2} \log \det \Sigma(q).$$

2.25 Convergence of Markov Chains

2.25.1 Spectral Gap

Theorem 2.25.1. *Let λ_2 be the second largest eigenvalue. Then*

$$\mathrm{Var}_\mu(K^n f) \leq |\lambda_2|^{2n} \mathrm{Var}_\mu(f).$$

2.25.2 Non-reversible Chains

Remark 2.25.1. For non-reversible chains:

$$\chi^2(\nu K^n || \mu) \leq C|\lambda_2|^{2n} \chi^2(\nu||\mu).$$

Remark 2.25.2. In continuous time this is called *hypocoercivity*.

2.25.2.1 Poincaré Inequality

Definition 2.25.1. A distribution μ satisfies a Poincaré inequality if

$$\langle g, \mathcal{L} g \rangle_\mu \geq c \mathrm{Var}_\mu(g).$$

Remark 2.25.3. Implies exponential convergence rate e^{-2ct}.

2.25.3 Contraction in Divergences

Definition 2.25.2. A kernel K is contractive if

$$D_f(\nu K||\mu) \leq (1-\kappa) D_f(\nu||\mu).$$

Remark 2.25.4. Implies geometric convergence.

2.25.4 Mixing Time

Definition 2.25.3.

$$t_{\mathrm{mix}}(\varepsilon) = \min\{t : \sup_{\mu_0} \mathrm{TV}(\mu_0 K^t, \mu) \leq \varepsilon\}.$$

Definition 2.25.4. Relaxation time:

$$t_{\mathrm{rel}} = \frac{1}{1-|\lambda_2|}.$$

2.25.5 Conductance

Definition 2.25.5. For $S \subset \Omega$:

$$\Phi(S) = \frac{Q(S, S^c)}{\mu(S)}, \quad Q(S, S^c) = \sum_{x \in S, y \in S^c} \mu(x) P(x, y).$$

Definition 2.25.6.

$$\Phi_* = \min_{\mu(S) \leq 1/2} \Phi(S).$$

Theorem 2.25.2.

$$t_{\text{mix}}(1/4) \geq \frac{1}{4\Phi_*}.$$

2.25.6 Asymptotic Variance

Definition 2.25.7.

$$\sigma_f^2 = \text{Var}_\mu(f(X_0)) + 2 \sum_{k=1}^{\infty} \text{Cov}_\mu(f(X_0), f(X_k)).$$

Theorem 2.25.3.

$$\frac{1}{N} \sum_{k=0}^{N-1} f(X_k) \to \mathbb{E}_\mu[f]$$

and

$$\text{Var}\left(\frac{1}{N} \sum f(X_k)\right) \sim \frac{\sigma_f^2}{N}.$$

2.25.7 Practical Diagnostics

- Trace plots (check stationarity)
- Autocorrelation:
$$\rho_\ell = \frac{\text{Cov}(f(X_0), f(X_\ell))}{\text{Var}(f)}$$
- Effective sample size:
$$N_{\text{eff}} = \frac{N}{1 + 2 \sum_{\ell \geq 1} \rho_\ell}$$

Discussion

- Single long chain vs multiple shorter chains:
  - single chain: cheaper burn-in
  - multiple chains: better exploration of multimodal distributions

2.26 Tempering and Particle Methods

2.26.1 Multimodality and Slow Mixing

Multimodal target distributions pose a fundamental challenge for MCMC.

- Probability mass is concentrated in separated regions (modes).
- Transitions between modes require crossing *energy barriers*.
- Local proposals (e.g., random walk, Langevin) rarely cross these barriers.

Remark 2.26.1. For distributions of the form $p(x) \propto e^{-f(x)}$, the time to transition between modes is often *exponential* in the barrier height:

$$\text{mixing time} \sim \exp(\Delta f).$$

This is known as *metastability*.

2.26.2 Temperature Scaling

Introduce a *temperature parameter* $T > 0$:

$$p_T(x) \propto e^{-f(x)/T} = e^{-\beta f(x)}, \quad \beta = \frac{1}{T}.$$

- High temperature ($T \gg 1$, $\beta \ll 1$): distribution is *flattened*.
- Low temperature ($T \ll 1$, $\beta \gg 1$): distribution concentrates near modes.

Remark 2.26.2. In Langevin dynamics:

$$dX_t = -\nabla V(X_t)\, dt + \sqrt{2T}\, dB_t,$$

temperature controls the tradeoff between gradient (drift) and noise.

2.26.3 Simulated Annealing

Simulated annealing gradually lowers temperature:

$$T_1 > T_2 > \cdots \to 0.$$

- Initially explores broadly.
- Eventually concentrates near global minima.

Remark 2.26.3. Once modes separate at low temperature, the chain becomes *trapped*: probability mass cannot easily move between modes.

2.26.4 Simulated Tempering

Introduce temperature as an auxiliary variable:

$$(\ell, x) \in \{1, \dots, L\} \times \Omega.$$

Define target:

$$\pi(\ell, x) \propto w_\ell \, p_\ell(x), \quad p_\ell(x) \propto e^{-f(x)/T_\ell}.$$

Algorithm:

- Alternate:
  - Update x using MCMC at fixed temperature ℓ.
  - Propose $\ell \to \ell \pm 1$ with MH acceptance:

$$\alpha = \min\left\{ \frac{w_{\ell'} p_{\ell'}(x)}{w_\ell p_\ell(x)}, 1 \right\}.$$

Remark 2.26.4. Drawback: requires knowledge (or estimation) of normalizing constants for p_ℓ.

2.26.5 Parallel Tempering (Replica Exchange)

Run multiple chains at different temperatures:

$$(x_1, \dots, x_L), \quad x_\ell \sim p_\ell.$$

Target:

$$\pi(x_1, \dots, x_L) = \prod_{\ell=1}^{L} p_\ell(x_\ell).$$

Swap proposal:

$$(x_i, x_{i+1}) \to (x_{i+1}, x_i),$$

with acceptance probability

$$\alpha = \min\left\{ \frac{p_i(x_{i+1}) p_{i+1}(x_i)}{p_i(x_i) p_{i+1}(x_{i+1})}, 1 \right\}.$$

Remark 2.26.5. Advantages:

- No need for normalizing constants.
- High-temperature chains help low-temperature chains escape modes.

2.26.6 Sequential Monte Carlo (SMC)

Construct a sequence of distributions:

$$p_k(x) \propto \tilde{p}_k(x), \quad k = 1, \ldots, L,$$

typically interpolating between easy and target distributions.

Algorithm 11 Sequential Monte Carlo

1: Sample $X_1^{(i)} \sim p_1$
2: **for** $k = 2$ to L **do**
3: Compute weights:

$$w_k^{(i)} \propto \frac{\tilde{p}_k(X_{k-1}^{(i)})}{\tilde{p}_{k-1}(X_{k-1}^{(i)})}$$

4: Resample particles according to $w_k^{(i)}$
5: Move particles: $X_k^{(i)} \sim K_k(X_{k-1}^{(i)}, \cdot)$
6: **end for**
7: Output $\{X_L^{(i)}\}$

Remark 2.26.6. SMC combines:

- Importance sampling (weights)
- Resampling (to prevent degeneracy)
- MCMC moves (to diversify particles)

2.26.7 Annealed Importance Sampling (AIS)

A simplified version of SMC without resampling.

Algorithm 12 Annealed Importance Sampling

1: Sample $X_1^{(i)} \sim p_1$
2: **for** $k = 2$ to L **do**
3: Sample $X_k^{(i)} \sim K_k(X_{k-1}^{(i)}, \cdot)$
4: **end for**
5: Compute weights:

$$\tilde{w}^{(i)} = \prod_{k=2}^{L} \frac{\tilde{p}_k(X_k^{(i)})}{\tilde{p}_{k-1}(X_k^{(i)})}$$

6: Estimate:

$$\mathbb{E}_{p_L}[f(X)] \approx \frac{\sum_i \tilde{w}^{(i)} f(X_L^{(i)})}{\sum_i \tilde{w}^{(i)}}$$

Remark 2.26.7. AIS is especially useful for:

- Estimating partition functions
- Bridging between simple and complex distributions

Key Insight

Tempering and particle methods address the same core issue:

Exploration vs. concentration tradeoff

- High temperature ⇒ exploration
- Low temperature ⇒ accurate sampling
- Tempering methods: move between temperatures
- Particle methods: propagate distributions across temperatures

2.27 Variational Methods

2.27.1 MCMC vs Variational Inference

Two paradigms for approximate inference:

- **MCMC:**
  - Constructs a Markov chain with stationary distribution p
  - Asymptotically exact
  - Slow mixing; hard to diagnose convergence
- **Variational methods:**
  - Solve an optimization problem over distributions
  - Fast and scalable
  - Approximate; may converge to local optima

Remark 2.27.1. MCMC: accuracy ↑, speed ↓
Variational inference: speed ↑, accuracy ↓ (approximation bias)

2.27.2 Gibbs Variational Principle

Let

$$p(x) = \frac{e^{f(x)}}{Z}, \quad Z = \int e^{f(x)} dx.$$

Lemma 2.27.1 (Gibbs variational principle)**.**

$$\ln Z = \max_q \left\{ H(q) + \mathbb{E}_{x\sim q}[f(x)] \right\}.$$

The maximum is attained at $q = p$.

Proof. Compute:

$$\mathrm{KL}(q\|p) = \int q(x) \log \frac{q(x)}{p(x)} dx = \int q(x) \log q(x) dx - \int q(x) f(x) dx + \log Z.$$

Rearrange:

$$\log Z = H(q) + \mathbb{E}_q[f(x)] + \mathrm{KL}(q\|p).$$

Since $\mathrm{KL}(q\|p) \geq 0$,

$$\log Z \geq H(q) + \mathbb{E}_q[f(x)],$$

with equality iff $q = p$. □

2.27.3 Variational Relaxation

We cannot optimize over all distributions q. Restrict to a class $\mathcal{Q}$:

$$\log Z \geq \max_{q\in\mathcal{Q}} \left\{ H(q) + \mathbb{E}_q[f(x)] \right\}.$$

Remark 2.27.2. Tradeoff:

- Larger $\mathcal{Q}$: better approximation, harder optimization
- Smaller $\mathcal{Q}$: easier optimization, worse approximation

2.27.4 KL Projection Interpretation

Using the identity:

$$\log Z = \mathrm{KL}(q\|p) + H(q) + \mathbb{E}_q[f(x)],$$

we obtain:

$$\arg\min_{q\in\mathcal{Q}} \mathrm{KL}(q\|p) = \arg\max_{q\in\mathcal{Q}} \left\{ H(q) + \mathbb{E}_q[f(x)] \right\}.$$

Remark 2.27.3. Variational inference = **projection of p onto $\mathcal{Q}$ in KL divergence**.

2.27.5 Two Use Cases

Partition function approximation:

$$\log Z \geq \max_{q \in \mathcal{Q}} \{H(q) + \mathbb{E}_q[f(x)]\}.$$

Distribution approximation:

$$q^* = \arg\min_{q \in \mathcal{Q}} \mathrm{KL}(q \| p).$$

For latent-variable models:

$$p(x) = \int p(x, z) dz,$$

we obtain:

$$\log p(x) = \mathrm{KL}(q(z) \| p(z|x)) + H(q) + \mathbb{E}_q[\log p(x, z)].$$

2.27.6 Evidence Lower Bound (ELBO)

Definition 2.27.1.

$$\mathrm{ELBO}(q) = H(q) + \mathbb{E}_{z \sim q}[\log p(x, z)].$$

$$\log p(x) \geq \mathrm{ELBO}(q).$$

Remark 2.27.4. Maximizing ELBO is equivalent to minimizing:

$$\mathrm{KL}(q(z) \| p(z|x)).$$

2.27.7 Mean-Field Variational Inference

Assume factorization:

$$q(x_1, \ldots, x_n) = \prod_{i=1}^{n} q_i(x_i).$$

Optimize coordinate-wise:

$$q_i^*(x_i) \propto \exp\left(\mathbb{E}_{q_{-i}}[f(x)]\right).$$

Algorithm 13 Coordinate Ascent Variational Inference (CAVI)

1: Initialize $q_1, \ldots, q_n$
2: **repeat**
3: **for** $i = 1, \ldots, n$ **do**
4: $$q_i(x_i) \propto \exp\left(\mathbb{E}_{q_{-i}}[f(x)]\right)$$
5: **end for**
6: **until** convergence

2.27.8 Approximate Likelihood-Based Learning

For latent-variable models $p_\theta(x, z)$:

$$\log p_\theta(x) = \max_q \{H(q) + \mathbb{E}_q[\log p_\theta(x, z)]\}.$$

Instead optimize:

$$\max_\theta \max_q \sum_{i=1}^N [H(q_i) + \mathbb{E}_{z\sim q_i} \log p_\theta(x_i, z)].$$

2.27.9 Amortized Variational Inference

Parameterize:

$$q_\phi(z|x).$$

Objective:

$$\max_{\theta,\phi} \sum_{i=1}^N \mathbb{E}_{z\sim q_\phi(z|x_i)} \left[\log \frac{p_\theta(x_i, z)}{q_\phi(z|x_i)}\right].$$

2.27.10 Variational Autoencoder (VAE)

- Encoder: $q_\phi(z|x)$
- Decoder: $p_\theta(x|z)$

ELBO:

$$\mathbb{E}_{z\sim q_\phi(z|x)}[\log p_\theta(x|z)] - \mathrm{KL}(q_\phi(z|x)\|p(z)).$$

Remark 2.27.5.
- First term: reconstruction
- Second term: regularization toward prior

2.27.11 Gradient Estimators

Score-function (REINFORCE):

$$\nabla_\phi \mathbb{E}_{z\sim q_\phi}[f(z)] = \mathbb{E}_{z\sim q_\phi}[f(z)\nabla_\phi \log q_\phi(z)].$$

Reparameterization trick:

$$z = g(x, \xi, \phi), \quad \xi \sim p(\xi),$$
$$\nabla_\phi \mathbb{E}_{z\sim q_\phi}[f(z)] = \mathbb{E}_\xi[\nabla_\phi f(g(x, \xi, \phi))].$$

Example 2.27.1.

$$z \sim \mathcal{N}(\mu_\phi(x), \Sigma_\phi(x)) \quad \Rightarrow \quad z = \mu_\phi(x) + \Sigma_\phi^{1/2}(x)\,\xi.$$

Remark 2.27.6.
- REINFORCE: general but high variance
- Reparameterization: lower variance but requires differentiability

2.27.12 Expectation Maximization (EM)

Algorithm 14 Expectation Maximization

1: **repeat**
2: **E-step:** $q(z) \leftarrow p_\theta(z|x)$
3: **M-step:** $\theta \leftarrow \arg\max_\theta \mathbb{E}_q[\log p_\theta(x, z)]$
4: **until** convergence

Remark 2.27.7. EM is a special case of variational inference with $\mathcal{Q}$ equal to all distributions.

2.27.13 Key Insight

Variational inference replaces:

$$\text{sampling} \quad \longrightarrow \quad \text{optimization.}$$

- MCMC: asymptotically exact, expensive
- Variational: approximate, scalable

2.27.14 Expectation Maximization (EM)

We consider latent-variable models $p_\theta(x, z)$ with observed data $x_1, \ldots, x_n$.

$$\max_\theta \sum_{i=1}^n \log p_\theta(x_i) = \max_\theta \sum_{i=1}^n \log \int p_\theta(x_i, z) dz.$$

Using the variational bound,

$$\log p_\theta(x_i) \geq \max_{q_i} \left[H(q_i) + \mathbb{E}_{z \sim q_i} \log p_\theta(x_i, z) \right].$$

Algorithm 15 Expectation Maximization (EM)

1: Initialize θ
2: **repeat**
3: **E-step:**

$$q_i(z) \leftarrow p_\theta(z|x_i)$$

4: **M-step:**

$$\theta \leftarrow \arg\max_\theta \sum_{i=1}^n \mathbb{E}_{z \sim q_i}[\log p_\theta(x_i, z)]$$

5: **until** convergence

Remark 2.27.8. EM alternates between:

- computing posteriors over latent variables
- maximizing expected complete-data log-likelihood

2.27.15 Example: Mixture of Gaussians

Model:

$$Z_m \sim \text{Categorical}(\pi), \quad X_m \sim \mathcal{N}(\mu_{Z_m}, \Sigma_{Z_m}).$$

E-step: posterior responsibilities

$$z_{mj} := p(Z_m = j | X_m = x_m) = \frac{\pi_j \mathcal{N}(x_m; \mu_j, \Sigma_j)}{\sum_k \pi_k \mathcal{N}(x_m; \mu_k, \Sigma_k)}.$$

M-step:

$$\pi_j \leftarrow \frac{1}{n} \sum_{m=1}^{n} z_{mj},$$

$$\mu_j \leftarrow \frac{\sum_{m=1}^{n} z_{mj} x_m}{\sum_{m=1}^{n} z_{mj}},$$

$$\Sigma_j \leftarrow \frac{\sum_{m=1}^{n} z_{mj} (x_m - \mu_j)(x_m - \mu_j)^\top}{\sum_{m=1}^{n} z_{mj}}.$$

2.27.16 Outer Relaxation

Instead of optimizing over global distributions $q(x)$, optimize over local marginals.

Let $p(x) \propto \exp\left(\sum_C \phi_C(x_C)\right)$.

Define pseudo-marginals $\{q_C\}$:

$$\mathbb{E}_q \left[\sum_C \phi_C(x_C) \right] = \sum_C \mathbb{E}_{q_C}[\phi_C(x_C)].$$

Relaxation:

$$\log Z \approx \max_{\{q_C\} \in \mathcal{Q}} \left[H(q) + \sum_C \mathbb{E}_{q_C}[\phi_C(x_C)] \right].$$

Remark 2.27.9. Constraints:

- consistency between marginals
- $q_C(x_C) \geq 0$, $\sum q_C = 1$

2.27.17 Marginal Polytope vs Local Polytope

- **Marginal polytope:** exact set of realizable marginals (intractable)
- **Local polytope:** enforce only pairwise consistency

Remark 2.27.10. Membership in marginal polytope is NP-hard.

2.27.18 Tree Case (Exact)

Lemma 2.27.2. *If the factor graph is a tree, local consistency implies global consistency.*

$$q(x) = \frac{\prod_{a \in F} q_a(x_a)}{\prod_{i \in V} q_i(x_i)^{d_i - 1}}.$$

2.27.19 Bethe Entropy

Lemma 2.27.3. *For tree-structured graphs,*

$$H(q) = \sum_{a \in F} H(q_a) - \sum_{i \in V} (d_i - 1) H(q_i).$$

Definition 2.27.2. The RHS defines the **Bethe entropy**:

$$H_{\text{Bethe}}(\{q_a\}) := \sum_a H(q_a) - \sum_i (d_i - 1) H(q_i).$$

2.27.20 Bethe Free Energy

Optimization problem:

$$\max_{\{q_a\} \in \mathcal{Q}} \left[H_{\text{Bethe}}(\{q_a\}) + \sum_a \mathbb{E}_{q_a}[\phi_a(x_a)] \right].$$

Remark 2.27.11.

- Exact on trees
- Approximate on loopy graphs
- Not necessarily convex

Note. Stationary points correspond to fixed points of loopy belief propagation.

2.27.21 KL Projections

Two types:

- **I-projection:**

  $$\min_{q \in \mathcal{Q}} \mathrm{KL}(q \| p)$$

  underestimates variance

- **M-projection:**

$$\min_{q \in \mathcal{Q}} \mathrm{KL}(p \| q)$$

  matches moments

2.27.22 M-Projection for Exponential Families

Let

$$q_\theta(x) = \frac{\exp(\langle \theta, \phi(x) \rangle)}{Z(\theta)}.$$

Lemma 2.27.4 (Moment matching)**.**

$$\arg\min_\theta \mathrm{KL}(p \| q_\theta) = \theta^* \quad s.t. \quad \mathbb{E}_{q_{\theta^*}}[\phi(x)] = \mathbb{E}_p[\phi(x)].$$

2.27.23 Belief Propagation as Variational Inference

Messages correspond to approximate marginals:

$$\nu_{i \to a}(x_i) \approx \text{marginal of } x_i.$$

Updates correspond to projections onto exponential family approximations.

Remark 2.27.12. Variational view:

- BP = coordinate ascent on Bethe free energy
- Loopy BP = approximate inference via Bethe relaxation

2.27.24 Example: Bayesian Linear Model

$$y = Ax + w, \quad w \sim \mathcal{N}(0, \sigma^2 I).$$

Posterior:

$$p(x|y) \propto \exp\left(-\frac{1}{2\sigma^2}\|y - Ax\|^2\right) p_0(x).$$

If

$$p_0(x) \propto e^{-\beta \|x\|_1},$$

then MAP becomes:

$$\min_x \|y - Ax\|^2 + \lambda \|x\|_1,$$

i.e., LASSO.

2.27.25 Approximate Message Passing (AMP)

To obtain tractable updates:

- Assume large random matrix A
- Use Gaussian approximations for messages

Remark 2.27.13. AMP emerges as a simplification of belief propagation under high-dimensional limits.

Chapter 3

Modern Deep Generative Models

3.1 Overview of Deep Generative Modeling

Goal: learn a distribution $p_\theta(x)$ from samples $\{x_i\}_{i=1}^N$ such that we can generate new samples.

Two broad paradigms:

- **Likelihood-based methods:**
  - Energy-based models (EBM)
  - Variational autoencoders (VAE)
  - Normalizing flows
- **Non-likelihood-based methods:**
  - Generative adversarial networks (GAN)
  - Noise contrastive estimation (NCE)
  - Score-based / diffusion models

Remark 3.1.1. Likelihood-based methods require evaluating or approximating $p_\theta(x)$, often needing inference (MCMC or variational).

3.2 Generative Modeling via Transformations

Let $z \sim p_0(z)$ be a simple distribution (e.g., $\mathcal{N}(0, I)$), and define

$$x = g_\theta(z).$$

Then p_θ is the pushforward:

$$p_\theta = (g_\theta)_{\#} p_0.$$

If g_θ is invertible:

$$p_\theta(x) = p_0(g_\theta^{-1}(x)) \left|\det Dg_\theta^{-1}(x)\right|.$$

Remark 3.2.1. This is the foundation of **normalizing flows**.

3.3 Normalizing Flows

3.3.1 Definition

Let $z \sim p_0(z)$ and $x = g(z)$ with invertible g. Then:

$$\log p(x) = \log p_0(z) + \log \left|\det Dg^{-1}(x)\right|.$$

Remark 3.3.1. Key requirement: efficient computation of $\det Dg^{-1}(x)$.

3.3.2 Composition of Flows

Let

$$g = g_L \circ \cdots \circ g_1.$$

Then:

$$\log \left|\det Dg^{-1}(x)\right| = \sum_{\ell=1}^{L} \log \left|\det Df_\ell(x_\ell)\right|,$$

where $f_\ell = g_\ell^{-1}$.

Remark 3.3.2. Design principle: compose simple invertible transformations with tractable Jacobians.

3.4 Building Blocks for Flows

3.4.1 Affine Transformations

$$f(x) = Ax + b.$$

- $\det Df = \det A$
- Efficient if A is triangular
- Limited expressivity

3.4.2 Coordinate-wise Transformations

$$f(x) = (f_1(x_1), \ldots, f_d(x_d)).$$

$$\det Df = \prod_{i=1}^{d} f_i'(x_i).$$

Remark 3.4.1. Keeps coordinates independent $\Rightarrow$ limited modeling power.

3.5 Affine Coupling Layers

Partition variables:

$$x = (x_1, x_2).$$

Define:

$$z_1 = x_1, \quad z_2 = x_2 \odot \exp(s(x_1)) + t(x_1).$$

Jacobian is triangular:

$$\log|\det Df(x)| = \sum_i s_i(x_1).$$

Remark 3.5.1.

- Efficient determinant: $\mathcal{O}(d)$
- Used in NICE, RealNVP, Glow

3.6 Residual Flows

Lemma 3.6.1. *Let $f(x) = x + F(x)$ where F is L-Lipschitz with $L < 1$. Then f is bijective.*

Theorem 3.6.1 (Banach fixed point)**.** *If T is a contraction, then it has a unique fixed point.*

Remark 3.6.1. Residual flows ensure invertibility via contraction mappings.

3.7 Efficient Log-Determinant Computation

$$\log\det(I + DF) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \operatorname{Tr}((DF)^k).$$

3.7.1 Hutchinson Trace Estimator

Lemma 3.7.1. *For $v \sim \mathcal{N}(0, I)$:*

$$\operatorname{Tr}(A) = \mathbb{E}[v^\top A v].$$

Remark 3.7.1. Allows stochastic estimation of log-determinants in high dimensions.

3.8 Flows in Variational Inference

Let

$$z = f_\phi(u, x), \quad u \sim \mathcal{N}(0, I).$$

Then ELBO becomes:

$$\mathbb{E}_u \left[\log p_\theta(x, f_\phi(u, x)) - \log q_\phi(f_\phi(u, x)|x) \right].$$

Remark 3.8.1. Flows increase expressivity of variational distributions.

3.9 Pros and Cons of Normalizing Flows

Pros:

- Exact likelihood evaluation
- Efficient sampling
- Invertible mapping enables interpolation

Cons:

- Requires invertibility constraints
- Jacobian computation can be costly
- May struggle with topological mismatch

3.10 Continuous-Time Flows (Neural ODEs)

Define dynamics:

$$\frac{dx(t)}{dt} = F(x(t), t), \quad x(0) = x_0.$$

Theorem 3.10.1 (Picard–Lindelöf)**.** *If F is Lipschitz, then the ODE has a unique solution.*

Remark 3.10.1. Defines a continuous flow:

$$x(T) = f_T(x_0).$$

3.11 Density Evolution

Change of variables:

$$\frac{d}{dt} \log p_t(x) = -\nabla \cdot F(x, t).$$

Remark 3.11.1. Leads to the **continuity equation**.

3.12 Score-Based Modeling

3.12.1 Score Function

Definition 3.12.1. The score function is:

$$s(x) = \nabla \log p(x).$$

3.12.2 Score Matching

$$L(\theta) = \mathbb{E}_{x\sim p}\|\nabla \log p(x) - s_\theta(x)\|^2.$$

Remark 3.12.1. Avoids computing normalization constant.

3.12.3 Integration by Parts Trick

$$\mathbb{E}\|\nabla \log p(x) - s_\theta(x)\|^2 = \mathbb{E}\left[\|s_\theta(x)\|^2 + \mathrm{Tr}(\nabla s_\theta(x))\right].$$

3.13 Denoising Score Matching

Add noise:

$$y = x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I).$$

Proposition 3.13.1. *Optimal estimator satisfies:*

$$s(y) = \frac{1}{\sigma^2}(\mathbb{E}[x|y] - y).$$

Remark 3.13.1. Links score estimation to denoising.

3.14 Diffusion Models (Preview)

Forward process:

$$dx_t = \sqrt{2}dB_t.$$

Reverse process depends on score:

$$dx_t = \nabla \log p_t(x)dt + \sqrt{2}dB_t.$$

Remark 3.14.1. Learning the score allows sampling from p via reverse-time SDE.

3.15 Summary

- Normalizing flows: exact likelihood via invertible maps
- Neural ODEs: continuous-time flows
- Score-based models: learn $\nabla \log p$
- Diffusion: simulate reverse-time dynamics

Chapter 4

Modern Deep Generative Models (II): Diffusion and Score-Based Methods

4.1 Statistical Complexity of Score Matching

Definition 4.1.1. The **log-Sobolev constant** of a distribution Q on $\mathbb{R}^d$ is the smallest C_{LS} such that for all distributions P,

$$\mathrm{KL}(P\|Q) \le C_{\mathrm{LS}}\,\mathrm{FI}(P\|Q),$$

where FI is the Fisher divergence.

Remark 4.1.1.

- Stronger than Poincaré inequality
- Implies exponential convergence of Langevin diffusion:

$$\mathrm{KL}(P_t\|Q) \le e^{-t/C_{\mathrm{LS}}}\mathrm{KL}(P_0\|Q)$$

4.1.1 Statistical Efficiency of Score Matching

Theorem 4.1.1 (Forbes & Lauritzen, 2015)**.** *Let $X_1, \dots, X_N \sim P_{\theta^*}$. Then the score matching estimator*

$$\hat{\theta}_{\mathrm{SM}} = \arg\min_{\theta} \sum_{i=1}^{N} \left[\frac{1}{2}\|\nabla \log p_\theta(X_i)\|^2 + \mathrm{Tr}(\nabla^2 \log p_\theta(X_i))\right]$$

satisfies

$$\sqrt{N}(\hat{\theta}_{\mathrm{SM}} - \theta^*) \xrightarrow{d} \mathcal{N}(0, \Gamma_{\mathrm{SM}}).$$

Remark 4.1.2. Score matching is typically less statistically efficient than MLE.

Theorem 4.1.2 (Koehler et al., 2022)**.** *If P_{θ^*} satisfies a restricted Poincaré inequality, then*

$$\|\Gamma_{\mathrm{SM}}\| \lesssim C_P^2\,\|I(\theta)\|^{-1} \cdot \textit{(moment terms)}.$$

Remark 4.1.3. Statistical gap depends on smoothness and functional inequalities.

4.2 Score Matching on Discrete Spaces

Definition 4.2.1. Let $G = (V, E)$ be a graph. Define the discrete gradient:

$$\nabla f(v) = (f(w) - f(v))_{w \in N(v)}.$$

Example (hypercube $\{\pm 1\}^n$):

$$\nabla_i f(x) = f(x^{(i)}) - f(x),$$

where $x^{(i)}$ flips coordinate i.

4.3 Pseudo-Likelihood

Definition 4.3.1. The pseudo-likelihood objective:

$$L_P(q) = \sum_{i=1}^{d} \mathbb{E}_{X \sim p}[\log q(X_i | X_{-i})].$$

Remark 4.3.1.

- Maximized at $q = p$
- Avoids global normalization
- Used in Ising models and graphical models

4.3.1 Generalized Pseudo-Likelihood

Partition coordinates into blocks $S_1, \ldots, S_k$:

$$L_P(q) = \sum_{i=1}^{k} \mathbb{E}_{X \sim p}[\log q(X_{S_i} | X_{S_i^c})].$$

4.4 Diffusion Models

4.4.1 Goal

Learn $p(x)$ and generate samples by transforming noise into data.

4.4.2 Forward Process

Define an SDE:

$$dx_t = f(x_t, t)dt + G(x_t, t)dW_t.$$

This gradually transforms data into noise.

4.4.3 Reverse Process

Theorem 4.4.1 (Anderson, 1982)**.** *The reverse-time SDE is:*

$$dy_t = \left[f(y_t, t) - \nabla \cdot (GG^\top p_t)(y_t)/p_t(y_t) \right] dt + G(y_t, t) d\widetilde{W}_t.$$

Remark 4.4.1. Reverse process depends on score $\nabla \log p_t(x)$.

4.4.4 Reverse Brownian Motion

$$dy_t = f(y_t, t)dt + G(y_t, t)d\widetilde{W}_t$$

is Brownian motion run backwards in time.

4.5 Score-Based Diffusion Framework

1. Define forward SDE:
$$dx_t = f(x_t, t)dt + g(t)dW_t$$
2. Learn score $s_\theta(x, t) \approx \nabla \log p_t(x)$
3. Simulate reverse SDE:
$$dx_t = \left[f(x_t, t) - g(t)^2 s_\theta(x_t, t) \right] dt + g(t)d\widetilde{W}_t$$

4.6 Variance Exploding (VE)

Forward process:

$$dx_t = g(t)dW_t$$

$$x_t = x_0 + \sigma(t)\epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

Variance increases over time.

4.7 Variance Preserving (VP)

$$dx_t = -\frac{1}{2}\beta(t)x_t dt + \sqrt{\beta(t)}dW_t$$

Remark 4.7.1. Ornstein–Uhlenbeck process.

4.8 Discrete-Time Diffusion (DDPM)

Forward:

$$x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{1-\alpha_t}\epsilon$$

Reverse:

$$p(x_{t-1}|x_t) \approx \mathcal{N}(\mu_\theta(x_t, t), \Sigma_t).$$

4.9 Connection to VAE

Remark 4.9.1. Diffusion models can be interpreted as hierarchical VAEs:

$$p(x_0, \ldots, x_T) = p(x_T) \prod_{t=1}^{T} p(x_{t-1}|x_t).$$

Training objective resembles ELBO:

$$\sum_t \mathbb{E}[\|\epsilon - \epsilon_\theta(x_t, t)\|^2].$$

4.10 Challenges with SDEs

- Brownian motion is not smooth
- Step size must be very small:

$$h = \mathcal{O}(1/d)$$

4.11 Probability Flow ODE

We can replace SDE with deterministic ODE:

$$dx_t = \left[f(x_t, t) - g(t)^2 \nabla \log p_t(x_t) \right] dt.$$

Remark 4.11.1.

- Same marginals as SDE
- Faster sampling (no noise)
- Sometimes less stable

4.12 Fokker–Planck Equation

Density evolution:

$$\frac{\partial p_t}{\partial t} = -\nabla \cdot (f p_t) + \frac{1}{2} \nabla^2 (G G^\top p_t).$$

4.13 Summary of Diffusion Models

- Learn score function $\nabla \log p_t(x)$
- Simulate reverse-time process
- Two main variants:
  - Variance exploding (VE)
  - Variance preserving (VP / DDPM)
- Alternative: probability flow ODE

4.14 Score vs Likelihood-Based Learning

- Score matching:
  - avoids normalization constant
  - less statistically efficient
- MLE:
  - statistically optimal
  - computationally hard

Remark 4.14.1. Functional inequalities (Poincaré, log-Sobolev) govern both:

- mixing of Markov chains
- statistical efficiency of learning

4.15 Probability Flow ODE and Sampling Refinements

4.15.1 Probability Flow ODE

Definition 4.15.1. Given the forward SDE

$$dx_t = f(x_t, t)\,dt + g(t)\,dW_t,$$

the **probability flow ODE** is

$$dx_t = \Big[f(x_t, t) - g(t)^2 \nabla \log p_t(x_t)\Big]\,dt.$$

Remark 4.15.1.

- Shares the same marginal distributions as the SDE.
- Deterministic $\Rightarrow$ faster sampling (no stochastic noise).

- Can compute likelihood via change-of-variables.
- May be less stable than SDE sampling.

4.15.2 Predictor–Corrector Methods

- **Predictor (P):** simulate reverse SDE/ODE step:

$$x_{t+h} \approx \text{reverse dynamics}$$

- **Corrector (C):** apply MCMC (Langevin step):

$$x \leftarrow x + h\nabla \log p_t(x) + \sqrt{2h}\,\xi, \quad \xi \sim \mathcal{N}(0, I).$$

- **PC scheme:** alternate P and C steps.

Remark 4.15.2. Corrector ensures local equilibrium; predictor moves across time.

4.16 Architecture and Implementation

4.16.1 Score Network

- Typically parameterized by a neural network $s_\theta(x, t)$
- For images: **U-Net architecture**
- Multi-scale structure critical for denoising

4.16.2 Practical Considerations

- Higher-order solvers (e.g., Heun)
- Noise schedule design
- Loss weighting across time
- Conditional diffusion models: learn $p(x|z)$
- Latent diffusion (VAE + diffusion)

4.17 Generalizing Diffusion Models

4.17.1 Observation Process View

Let $X \sim P$ be data. Define noisy observations:

$$Y_t = X + \text{noise}$$

- As $t \to \infty$: Y_t becomes pure noise
- As $t \to 0$: recover original data

4.17.2 Stochastic Localization

$$p(X \mid Y_t)$$

defines evolving distributions that interpolate between prior and data.

4.18 Diffusion on Discrete Spaces

- Embed discrete data into continuous space
- Masking / token corruption (language models)
- Define discrete Markov processes directly

4.18.1 Continuous-Time Markov Chain

Generator matrix A:

$$A_{xy} \geq 0, \quad A_{xx} = -\sum_{y \neq x} A_{xy}.$$

Generator:

$$\mathcal{L}f(x) = \sum_{y \neq x} A_{xy}(f(y) - f(x)).$$

Remark 4.18.1. Defines jump process with exponential waiting times.

4.19 Reversing Markov Processes

Theorem 4.19.1. *Let (P_t) be a Markov process with generator A and marginals p_t. The reverse process has generator*

$$B_t(x, y) = A_{yx} \frac{p_t(y)}{p_t(x)}, \quad x \neq y.$$

Remark 4.19.1. Discrete analogue of reverse SDE.

4.19.1 Example: Bit-Flip Process

On $\{-1, 1\}^n$:

- Forward: flip each coordinate randomly
- Reverse: depends on ratio $\frac{p_t(x^{(i)})}{p_t(x)}$

4.20 Stochastic Interpolants

Definition 4.20.1. A **stochastic interpolant** between distributions ρ_0, ρ_1 is:

$$x_t = I(t, x_0, x_1) + \gamma(t)z, \quad z \sim \mathcal{N}(0, I).$$

Example:

$$x_t = (1-t)x_0 + tx_1 + \sqrt{2t(1-t)}\, z.$$

Remark 4.20.1. Generalizes diffusion models to arbitrary endpoints.

4.20.1 Dynamics

- ODE: $dx_t = b_t dt$
- SDE: $dx_t = (b_t + s_t)dt + \sqrt{2\epsilon}dW_t$

4.21 Schrödinger Bridge

4.21.1 Problem

Find a stochastic process matching:

$$X_0 \sim \rho_0, \quad X_T \sim \rho_1$$

while minimizing KL divergence to a reference process.

Definition 4.21.1.

$$\pi^* = \arg\min_{\pi} \mathrm{KL}(\pi \| p_{\mathrm{ref}})$$

subject to endpoint constraints.

4.21.2 Connection to Optimal Transport

- Equivalent to entropy-regularized OT
- Balances transport cost and entropy

4.22 Iterative Proportional Fitting (IPF)

- Alternate between matching forward/backward marginals
- Converges to Schrödinger bridge solution

Remark 4.22.1. Key idea: alternating KL projections.

4.23 Diffusion vs Schrödinger Bridge

- Diffusion: fixed prior $\rightarrow$ data
- Schrödinger bridge: arbitrary $\rho_0 \rightarrow \rho_1$
- More flexible, but computationally heavier

4.24 Big Picture

- Score-based models learn $\nabla \log p_t$
- Reverse-time dynamics generate samples
- Diffusion = special case of stochastic interpolation
- Schrödinger bridge = optimal interpolation

4.25 Modern Deep Generative Modeling

4.25.1 Unifying View

We consider the problem:

$$\text{Learn } p_{\text{data}}(x) \text{ from samples.}$$

- **Likelihood-based:**
  - Energy-based models (RBM)
  - Variational Autoencoders (VAE)
  - Normalizing flows
- **Non-likelihood-based:**
  - GANs
  - Noise Contrastive Estimation (NCE)
  - Diffusion / score-based models

4.26 Generative Adversarial Networks (GANs)

4.26.1 Formulation

$$\min_{g \in \mathcal{G}} \max_{f \in \mathcal{F}} \left[\mathbb{E}_{x \sim p_{\text{data}}} \log f(x) + \mathbb{E}_{x \sim p_g} \log(1 - f(x)) \right]$$

- g: generator, pushes $z \sim p_z$ to $x = g(z)$
- f: discriminator, estimates $P(y = 1|x)$

4.26.2 Optimal Discriminator

$$f^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_g(x)}.$$

4.26.3 Divergence Interpretation

$$\min_g \; 2\,\text{JS}(p_g, p_{\text{data}})$$

Remark 4.26.1. GANs minimize Jensen–Shannon divergence.

4.26.4 Wasserstein GAN

$$\min_g \max_{f \in \text{Lip}(1)} \left[\mathbb{E}_{p_{\text{data}}} f(x) - \mathbb{E}_{p_g} f(x)\right]$$

- Approximates $W_1(p_{\text{data}}, p_g)$
- Requires Lipschitz constraint on f

Remark 4.26.2. More stable than standard GANs when supports are disjoint.

4.26.5 Practical Issues

- Mode collapse
- Training instability (min-max optimization)
- Sensitive to discriminator capacity

4.27 Noise Contrastive Estimation (NCE)

4.27.1 Idea

Reduce density estimation to classification:

$$\ell(f) = \sum_{i=1}^{N} \log f(x_i^{\text{data}}) + \sum_{j=1}^{M} \log(1 - f(x_j^{\text{noise}})).$$

- Data: $x \sim p_{\text{data}}$
- Noise: $x \sim q$

4.27.2 Optimal Classifier

$$f^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + kq(x)}.$$

Remark 4.27.1. Recovers density ratio $\frac{p_{\text{data}}}{q}$.

4.27.3 Key Insight

- Avoids computing partition function
- Choice of q critical (curse of dimensionality)

4.27.4 Telescoping Density Ratio Estimation

Construct intermediate distributions:

$$\frac{p_0}{p_n} = \frac{p_0}{p_1} \cdot \frac{p_1}{p_2} \cdots \frac{p_{n-1}}{p_n}.$$

4.28 Evaluation Metrics

4.28.1 Inception Score (IS)

$$\mathbb{E}_{x \sim p_g} \text{KL}(p(y|x) \| p(y))$$

- High if:
  - confident predictions (low entropy $p(y|x)$)
  - diverse outputs (high entropy $p(y)$)

Remark 4.28.1. Can be gamed (memorization).

4.28.2 Fréchet Inception Distance (FID)

$$W_2^2 = \|\mu - \mu'\|^2 + \text{Tr}(\Sigma + \Sigma' - 2(\Sigma\Sigma')^{1/2})$$

- Compares Gaussian approximations in feature space
- Measures both quality and diversity

4.29 Diffusion Models (Modern View)

4.29.1 Score-Based Perspective

$$s(x) = \nabla \log p(x)$$

- Learn score via score matching
- Sample via Langevin / reverse SDE

4.29.2 Forward / Reverse Processes

- Forward: add noise until $x_T \sim \mathcal{N}(0, I)$
- Reverse: remove noise using learned score

4.29.3 Two-Step View

1. Learn score $s_\theta(x, t)$
2. Simulate reverse dynamics to sample

4.30 Probability Flow and Sampling Refinements

- Reverse SDE (stochastic)
- Probability flow ODE (deterministic)
- Predictor-corrector methods

4.31 Score Matching on Discrete Spaces

Define discrete gradient:

$$\nabla f(x) = (f(x^{(i)}) - f(x))_{i=1}^{d}.$$

Alternative objective:

$$L_p(q) = \sum_{i=1}^{d} \mathbb{E}_{x\sim p} \log q(x_i | x_{-i}).$$

Remark 4.31.1. Pseudo-likelihood is discrete analogue of score matching.

4.32 Statistical Efficiency

4.32.1 MLE vs Score Matching

- MLE:
  - statistically optimal
  - computationally hard
- Score matching:

  - computationally efficient
  - statistically suboptimal

4.32.2 Functional Inequalities

- Poincaré inequality $\Rightarrow$ mixing
- Log-Sobolev inequality $\Rightarrow$ fast convergence

4.33 Final Big Picture

- Generative modeling = learning distributions
- Three main paradigms:
  - Optimization (VI, flows, GANs)
  - Sampling (MCMC, diffusion)
  - Hybrid (score-based methods)
- Tradeoffs:
  - Statistical efficiency vs computational tractability
  - Exactness vs scalability

4.34 Calibration and Conformal Prediction

4.34.1 Calibration

Goal: Ensure probabilistic predictions reflect true frequencies.

A predictor $f : \mathcal{X} \to [0, 1]$ is *calibrated* if

$$\mathbb{P}(Y = 1 \mid f(X) = p) = p \quad \forall p \in [0, 1].$$

Remark 4.34.1. Calibration is about *uncertainty correctness*, not accuracy.

- Perfect calibration but poor accuracy: always predict 1/2.
- Good accuracy but poor calibration: overconfident predictions (0 or 1).

4.34.2 Calibration in Deep Learning

- Overparameterized models $\Rightarrow$ overconfidence.
- Minimizing loss $\not\Rightarrow$ calibrated probabilities.

- Post-training (e.g., RLHF) can worsen calibration.

4.35 Proper Scoring Rules

A scoring rule S is *proper* if:

$$\arg\max_{\hat{p}} \mathbb{E}[S(\hat{p}, Y)] = p.$$

Examples:

- Log score: $S(p) = \log p$
- Brier score: $S(p) = -(1-p)^2$

Remark 4.35.1. Proper scoring rules incentivize truthful probabilities.

4.36 Measuring Calibration

4.36.1 Expected Calibration Error (ECE)

$$\mathrm{ECE}(f) = \mathbb{E}\Big[\, |\mathbb{E}[Y \mid f(X)] - f(X)| \,\Big].$$

$$\mathrm{ECE}_2(f) = \mathbb{E}\Big[\, (\mathbb{E}[Y \mid f(X)] - f(X))^2 \,\Big].$$

Remark 4.36.1. In practice, estimated via binning (reliability diagrams).

4.36.2 Decomposition of Error

$$\mathbb{E}[(f(X) - Y)^2] = \underbrace{\mathbb{E}[(f(X) - \mathbb{E}[Y|f(X)])^2]}_{\text{calibration error}} + \underbrace{\mathbb{E}[(\mathbb{E}[Y|f(X)] - Y)^2]}_{\text{intrinsic noise}}.$$

4.37 Calibration Algorithms

Define:

$$\mathrm{calib}(p) = \mathbb{P}(Y = 1 \mid f(X) = p).$$

Construct calibrated predictor:

$$\hat{f}(x) = \mathrm{calib}(f(x)).$$

Theorem 4.37.1. *$\hat{f}$ is calibrated and improves squared loss:*

$$L(\hat{f}) = L(f) - \mathrm{ECE}_2(f).$$

4.37.1 Practical Methods

- Binning / histogram calibration
- Isotonic regression
- Temperature scaling:

$$\hat{f}(x) = \frac{e^{\beta f(x)}}{1 + e^{\beta f(x)}}$$

4.38 Issues in Continuous Case

- Cannot directly estimate $\mathbb{P}(Y = 1 | f(X) = p)$
- ECE is discontinuous

Fixes:

- Binning
- Smoothing
- Distance to calibration:

$$d_{\text{CE}}(f) = \inf_{g \in \text{Cal}} \mathbb{E}|f(X) - g(X)|$$

4.39 Extensions

- Multicalibration (fairness across groups)
- Multiclass calibration
- Decision calibration (task-aware)

4.40 Conformal Prediction

4.40.1 Goal

Construct set-valued predictor $T(x)$ such that:

$$\mathbb{P}(Y \in T(X)) \geq 1 - \delta.$$

4.40.2 Nonconformity Score

Define $s(x, y)$ measuring prediction error.

Examples:

- Regression: $s(x, y) = |f(x) - y|$
- Classification: $s(x, y) = 1 - f(x, y)$

4.40.3 Algorithm

Given data $(x_i, y_i)_{i=1}^n$:

$$\tau = \min \left\{ t : \sum_{i=1}^{n} \mathbf{1}_{s(x_i,y_i)\leq t} \geq (1-\delta)(n+1) \right\}.$$

Output:

$$T(x) = \{y : s(x, y) \leq \tau\}.$$

4.40.4 Guarantee

Theorem 4.40.1. *For new (x, y):*

$$1 - \delta \leq \mathbb{P}(y \in T(x)) \leq 1 - \delta + \frac{1}{n+1}.$$

4.40.5 High-Probability Version

$$\tau = \text{quantile}_{1-\delta} + \sqrt{\frac{\ln(2/\gamma)}{2n}}$$

4.41 Key Insight

- Calibration $\Rightarrow$ correctness of probabilities
- Conformal prediction $\Rightarrow$ correctness of *sets*
- No distributional assumptions needed (only exchangeability)

4.42 Big Picture

- MCMC / VI: approximate distributions
- Calibration: fix uncertainty of predictions
- Conformal: wrap any model with guarantees